Cash Transfers, Gambling and Domestic Violence

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Preliminary draft. Comments welcome.

Abstract

We study whether the viral spread of online casino gambling increased domestic violence around Bolsa Família cash transfer paydays. We exploit the sudden popularity of the "jogo do tigrinho," a slot game that went viral in Brazil in mid-2023, and the Bolsa Família payment calendar, which assigns payment days by the last digit of each beneficiary's social identification number. Using daily health surveillance records and a triple difference-in-differences design, we find that after the game went viral, payments increased violence against women and minors in the household in municipalities with higher program coverage, with no effect on male victims. The effect is concentrated in physical violence and in male-on-female intimate partner violence. Our results suggest that online gambling may erode the protective role that higher program coverage played for children and intensify disputes between partners over the household's liquidity.

Keywords: Online Casino, Jogo do Tigrinho, Bolsa Família, Payday Effects, Intimate Partner Violence, Brazil

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1. Introduction

The unprecedented expansion of digital gambling in recent years has turned it into one of the new public concerns of our time, and a growing priority in public debate. Far from being an isolated trend in a few countries, online gambling has become a global public health challenge that threatens social well-being (World Health Organization, 2024). This growing concern contrasts with the scarce evidence on the potential negative consequences of digital gambling and on its influence on behavior. The gap is particularly marked when it comes to its potential impact on socioeconomic well-being, both for gamblers and for their families.

This paper studies whether the viral spread of online casino gambling increased domestic violence in Brazil around the days on which households receive cash transfer payments under the Bolsa Família program. We focus on the "jogo do tigrinho" (Fortune Tiger), a smartphone-based, virtual slot machine game that went viral in Brazil within a few months. We combine the sudden emergence of the game and its rapid adoption among low-income users with exogenous variation in payment timing generated by the administrative rule that sets the Bolsa Família payment calendar. This allows us to estimate the effect of liquidity shocks on intra-household violence during the peak months of the game's popularity.

By 2023, Brazil was already among the ten largest betting markets in the world. In mid-2023, the viral spread of Tigrinho reshaped this market. The game spread rapidly through influencer content on TikTok and WhatsApp, and Google searches for it grew by 170 percent month-over-month between April and August 2023. Unlike sports betting, which tends to attract younger bettors who place more informed wagers, a slot game requires no knowledge of odds or sports. Its minimalist design and simple rules led to disproportionate uptake among lower-income households, and in particular among beneficiaries of Bolsa Família, one of the largest cash transfer programs in the world.

The scale of the phenomenon is substantial. According to the Central Bank of Brazil, in August 2024 alone, 5 million people in Bolsa Família households—17 percent of all

beneficiaries—sent R$3 billion to betting companies via Pix, Brazil's instant payment system, with a median transfer of R$100 per person (Banco Central do Brasil, 2024). This amount is equivalent to roughly 20 percent of the R$14.1 billion the government transferred through the program that month. In response, Brazil's Supreme Court ordered in November 2024 that social assistance benefits be barred from use in online betting, a restriction the Ministry of Finance only regulated in October 2025.

These features—the sudden viral spread of the game and its close connection to Bolsa Família—provide a setting well suited for causal analysis. We match daily records of domestic violence cases from Brazil's health surveillance system (SINAN) with municipal-level administrative records on Bolsa Família beneficiaries, the program's monthly payment calendars, and a proxy for the game's popularity based on Google search intensity. Our triple difference-in-differences design compares domestic violence around paydays, before and after the game went viral, across municipalities with different levels of Bolsa Família coverage. We focus on the period between February 2022 and June 2024, before regulatory measures began to affect the online casino market.

The key input for the payday dimension comes from the Federal Government's Transparency Portal (Portal da Transparência), which reports the number of Bolsa Família beneficiaries in each of Brazil's 5,570 municipalities by the last digit of their social identification number (NIS). Because the program pays each beneficiary on a day determined by this last digit, these counts allow us to trace, day by day, how payments actually unfold over the monthly payment cycle in every municipality. The assignment rule does not depend on any beneficiary or household characteristic, so the resulting variation in payment timing is as good as random.

To measure exposure to payments, we exploit two sources of variation in payment timing: differences across municipalities in the number of beneficiaries assigned to each day of the payment window, and month-to-month shifts in the days on which that window opens and closes (Foley, 2011; Hsu, 2017; Carr and Packham, 2021). We summarize both in a single

treatment measure: the cumulative share of a municipality's beneficiaries who have already been paid on each day of the month. Our coefficient of interest captures how the effect of this measure on domestic violence changes after the game went viral, and how this change varies with Bolsa Família coverage.

Because Bolsa Família payments are concentrated in the second half of the month, in our preferred specification we include day-of-month fixed effects, ensuring that our estimates are not driven by recurring end-of-month patterns, such as the build-up of financial stress or tighter household budget constraints. We also include municipality, year-by-month, and day-of-week fixed effects, and alternative specifications add further municipal-level controls.

We find that, after the game went viral, Bolsa Família payments significantly increased the daily number of domestic violence cases against women and children in municipalities with higher program coverage. In contrast, we find no effect on domestic violence cases in which the victim is male. Event-study estimates show a discrete jump in the payday effect following the Tigrinho boom, with no evidence of differential trends in the preceding quarters. Before that, the relationship between Bolsa Família payments and domestic violence is generally positive but varies in magnitude for adults and minors and for municipalities with higher levels of coverage, a pattern we discuss in detail in the results section.

The effect amounts to a 3.5 percent increase relative to the mean for all women, 6 percent for girls, and 2.7 percent for adult women. The increase is concentrated in physical violence and intimate partner violence, and is driven by male perpetrators. Physical violence also increases among child victims.

Several mechanisms may link liquidity shocks and online gambling to domestic violence, in line with the literature on family violence (Straus et al., 1980; Johnson, 2009), economic models of addiction (Bernheim and Rangel, 2004), and failures of self-control (Loewenstein and O'Donoghue, 2007). One interpretation of our results builds on the upset-loss framework of Card and Dahl (2011), in which unexpected losses act as emotional cues that trigger violence. Although the odds of a slot game are stacked against the player, cognitive biases

such as the illusion of control (Langer and Roth, 1975; Griffiths, 1994) and the gambler's fallacy (Clotfelter and Cook, 1993) lead gamblers to expect a win. A losing bet may therefore be experienced as an upset loss and trigger domestic conflict in the same way.

Given the high opacity of gambling data, we cannot measure betting outcomes or their timing, and we cannot rule out other mechanisms. First, losses may be fully anticipated, and violence may instead result from the additional financial stress they create. Second, conflict may arise from tension within the couple, as one partner reproaches the other for gambling, especially when it is hidden. In a more extreme case, both spouses may agree to take part in online betting, but conflict may still be triggered by a dispute over the allocation of resources relative to other household needs.

Although our data do not allow us to test these mechanisms separately, SINAN records the relationship between victim and perpetrator and the place where the event occurred, which allows us to examine where the effect comes from. For violence against women, the effect is driven by cases perpetrated by a male cohabiting partner, whether a spouse or a de facto partner, and we find no effect for other perpetrators, such as relatives, co-workers or supervisors, or strangers. The effect is also concentrated in events that occur at home rather than in public spaces, at work, or at school. Within intimate partner violence, the increase is driven by current partners, not by former partners or non-cohabiting romantic partners.

This evidence does not identify which channel dominates, but it supports a condition that all of them require: that the conflict takes place between partners who share a home. Each mechanism relies on a feature of cohabitation. Financial stress and disputes over resources require a shared budget; reproach over gambling requires an ongoing relationship; and emotional cues from upset losses require that the gambler be in close contact with the victim when the loss occurs.

We subject our results to several placebo tests. We estimate the same specification on health-system outcomes that should not respond to the interaction between gambling and liquidity flows, randomly permute Bolsa Família coverage across municipalities, and assign

fake payment calendars. None of these exercises produces effects comparable to our main estimates. Our estimates are also unchanged when we control for the new benefit components that Bolsa Família introduced in June 2023. For months in which withdrawal records are available, we verify that our measure of cumulative payments closely tracks the timing of actual cash withdrawals. Finally, the results are robust to alternative sample definitions and to continuous versions of the treatment variables.

Our main limitation is that we do not observe gambling directly. Before betting was regulated in Brazil, online casinos operated from abroad and with little transparency, so no reliable measure of spending on these platforms exists for our period (Carvalho et al., 2026). Transaction data would not solve this problem, since payments to offshore operators cannot be reliably identified in them, and the Central Bank of Brazil has repeatedly denied requests for such data under Brazil's freedom-of-information law (LAI). We therefore rely on Google search intensity as a proxy for the game's popularity. We believe this proxy is appropriate: because the game was not available in the Google and Apple app stores, most users accessed it through web browsers, which makes online search activity a reasonable indicator of its popularity (Section 2).

To the best of our knowledge, this is the first paper to provide causal evidence on the effect of unregulated online casino gambling on domestic violence, and the first in Latin America to study the consequences of gambling beyond gamblers' own financial outcomes. We also show how these effects interact with the liquidity provided by a poverty-targeted cash transfer program.

Our paper contributes to three strands of literature. First, we contribute to the literature on the consequences of gambling. Recent studies document negative financial effects of gambling involvement in the United States (Baker et al., 2026) and in Brazil (Carvalho et al., 2026). Beyond financial outcomes, Matsuzawa and Arnesen (2024) extend the framework of Card and Dahl (2011) and show that the legalization of sports betting in the United States amplifies the effect of emotional cues on intimate partner violence. We extend this evidence

to unregulated online casino gambling rather than legal sports betting, to several types of family violence beyond intimate partner violence, and by exploring the interaction with the timing of payments from a large cash transfer program in Latin America.

Second, we contribute to the literature on cash transfers and household conflict. Transfers may reduce domestic violence by easing financial stress and strengthening women's bargaining position, or increase it when partners use violence to gain control over the new resources (Angelucci, 2008; Bobonis et al., 2013; Haushofer et al., 2019). Consistent with this theoretical ambiguity, the empirical evidence is mixed and depends on the outcome, program, design, and context (Buller et al., 2018; Hidrobo and Fernald, 2013; Hidrobo et al., 2016; Roy et al., 2019), including evidence for Bolsa Família itself (Litwin et al., 2019). We show that, even within the same program, the effect of payments differs between violence against adult women and violence against children. More importantly, we show that the balance between these channels is not fixed: the arrival of an easily accessible temptation good, online gambling, can amplify an adverse effect or offset a protective one, acting as a channel through which liquidity leaks out of the household.

Third, we contribute to the literature on the timing of benefit payments and crime. Payment schedules have been shown to affect crime (Foley, 2011) and domestic violence (Hsu, 2017; Carr and Packham, 2021; Masi and Santantonio, 2026), and our results provide evidence in a similar direction for Latin America. We also contribute methodologically. Hsu (2017) infers payment dates from state-level TANF calendars, common to all recipients within a state, while Carr and Packham (2021) identify effects off a single policy change: Illinois's 2010 switch to staggered SNAP issuance. Bolsa Família payment timing, by contrast, is set by the last digit of each beneficiary's NIS, an administrative rule assigned independently of household characteristics. This yields variation in payment timing that originates in an individual-level assignment rule and repeats every month throughout our sample period. Under this setting, we can control flexibly for day-of-month effects and separate Bolsa Família payments from generic end-of-month patterns, such as the timing of private paychecks.

This paper also contributes to the policy debate. In Latin America and in developing countries more broadly, cash transfers are a major source of income for a large share of households. If part of this liquidity is diverted toward gambling, the harms of gambling may be amplified precisely among vulnerable households, and part of the transfer's welfare gains may be lost. Our results suggest that the design of payment schemes should take into account the potential for problematic consumption around paydays. They also point to the value of gambling prevention policies for the population as a whole, and of violence prevention efforts targeted at women and children in beneficiary households.

These lessons extend beyond Brazil. Several countries in the region have only recently regulated online gambling or are still debating how to do so. Our results identify a risk that can be anticipated and factored into these regulatory frameworks: the interaction between the viral spread of online gambling and the timing of cash transfer payments. While we do not evaluate specific policies, our findings provide a diagnosis that motivates considering targeted tools, such as limits on the amount or timing of deposits, self-exclusion mechanisms, or alerts for cash transfer beneficiaries, as well as stronger enforcement against the illegal gambling market.

Our results also highlight the need to reduce the opacity of gambling data, for instance by requiring operators to report their activity so that the market can be monitored. Without such measures, the expansion of online gambling may generate broader and unintended consequences, affecting not only gamblers' finances but also the well-being of their families and the effectiveness of the social policies designed to protect them.

The rest of the paper is organized as follows. Section 2 describes the regulation of gambling in Brazil and the rise of the jogo do tigrinho. Section 3 presents the data sources and the construction of the sample. Section 4 describes the identification strategy. Section 5 presents the main findings. Section 6 explores heterogeneity in the effects, and Section 7 offers a battery of robustness checks. Finally, Section 8 concludes.

2. Institutional Background

2.1. Gambling Regulation in Brazil

Gambling has been largely prohibited in Brazil since 1946, when a decree-law banned casinos and most games of chance, leaving government-run lotteries as the main legal exception. In 2018, Law 13.756 created fixed-odds betting and gave the Ministry of Finance two years, extendable for two more, to regulate it. The deadline expired in December 2022 without regulation. Throughout our study period, therefore, no company held a federal license to offer online betting in Brazil. The online gambling market was thus unregulated, but not entirely illegal, and it comprised two distinct segments. Sports betting had been legalized by the 2018 law but, in the absence of regulation, operated in a legal gray area. Online casino games, such as the jogo do tigrinho, were not covered by that law and remained illegal. Operators, many of them incorporated abroad but targeting Brazilian customers, offered these games without a domestic legal entity or public registry, and routed payments through intermediaries registered under unrelated activities (Carvalho et al., 2026).[1]

Law 14.790 of December 2023 brought fixed-odds betting, including online casino games, under a federal licensing regime, and the Ministry of Finance issued its implementing rules in April and May 2024. These rules, however, applied only to licensed operators, and companies already active in Brazil had until December 2024 to comply. Companies could apply for a license from May 2024, and those that had not applied by the end of September 2024 were barred from operating, with their websites blocked from October 2024 onward. The regulated regime took full effect in January 2025. Our sample therefore ends before any regulatory measure affected the market.

The link between online gambling and Bolsa Família became a policy priority shortly

[1]Most of these payments were made through Pix, the instant payment system launched by the Central Bank of Brazil in November 2020. Pix allows free, real-time transfers between any two accounts and is now used almost universally in Brazil. By 2025, it accounted for about 97 percent of all deposits on betting platforms (Carvalho et al., 2026).

after our study period. After the Central Bank estimated that about one in six beneficiaries had sent money to betting companies in August 2024 (Banco Central do Brasil, 2024), the Supreme Court ordered the government to prevent the use of social benefits for betting. Since October 2025, licensed operators must check each bettor's tax identification number (CPF) against the list of Bolsa Família and BPC beneficiaries and deny them an account.[2] In September 2026, a provisional measure went further and banned fixed-odds betting, including online casino games, throughout the country (Brasil, 2026); it took effect immediately but requires congressional approval to become permanent.

2.2. The Jogo do Tigrinho

The jogo do tigrinho is the popular name of Fortune Tiger, a slot machine game released in 2022 by PG Soft, a developer based in Malta. As is common in the online casino industry, PG Soft does not offer its games to players directly. Instead, it licenses them to online casino operators through game aggregators, which bundle titles from many developers into a single catalog. In Brazil, the game was offered by operators without a federal license, including Brazilian groups.[3] Alongside these operators, a large number of clandestine websites offered unlicensed copies of the game with manipulated payouts, often as outright fraud schemes. The opacity of the market also gave rise to imitation apps. In September 2023, an unofficial imitation of the game, not affiliated with PG Soft, was the tenth most downloaded app in Brazil, alongside apps such as Facebook, PicPay, and Spotify, with 2.5 million downloads in a single month (Mobile Time, 2023).

The game spread through affiliate programs that paid influencers a commission on the deposits, or losses, of the players they referred. Many of them posted videos of large wins recorded on demonstration accounts, which simulate results without real money. Law en-

[2]Portaria SPA/MF No. 2.217 of 2025. In December 2025, the Supreme Court suspended the closure of existing accounts pending a conciliation hearing, while keeping the ban on new accounts.

[3]In its testimony to the Senate's parliamentary inquiry into online betting, one of the companies identified as responsible for spreading the game acknowledged offering it through an aggregator (Senado Federal, 2024).

forcement responded by targeting these promoters rather than the operators: from December 2023 onward, state police forces carried out several operations against influencers accused of advertising the game with fake wins, on charges that included fraud and money laundering.

Due to its illegal nature, players could not download the original game from official app stores. Until mid-2025, Google Play and the Apple App Store did not list real-money gambling apps for Brazil, so access to online betting was almost exclusively through mobile web browsers (Carvalho et al., 2026). The game runs inside the operator's website, and players typically reached it by searching for its name or by following an affiliate link. Internet searches for the game are therefore a natural indicator of interest in it, which is why we use Google search intensity to measure its popularity (Section 3).

Unfortunately, actual gambling on the jogo do tigrinho is unobservable in this study, not only because the data are not accessible, but because of the nature of the market itself. To our knowledge, and unlike sports betting, there is no way to measure how much users bet on the game, given that it operated in an unregulated market. The most promising source would be transaction-level Pix data, but even these would be incomplete. Until 2025, payments to unauthorized operators were routed through intermediaries registered under unrelated activities, so transfers to these platforms cannot be identified as bets. Even when they can, the type of game cannot: Carvalho et al. (2026), who observe individual Pix transfers, can classify operators as casino or sports betting only from 2025 onward, once authorized operators could be linked to their brands. Online casinos, which were mostly unauthorized, are therefore invisible in transaction data for our period.

3. Data and Sample

3.1. Domestic Violence: SINAN

Our outcomes come from the Notifiable Diseases Information System (Sistema de Informação de Agravos de Notificação, SINAN), administered by Brazil's Ministry of Health.

Health professionals in public and private facilities are required to notify every suspected or confirmed case of interpersonal or self-inflicted violence they attend. We use the case-level microdata for 2022–2024. Each record reports the date of the event, the type of violence, the characteristics of the victim, the relationship between victim and perpetrator, and the place where the event occurred. We assign each case to the victim's municipality of residence. Because notification does not depend on the victim filing a police report, SINAN captures cases that may never reach the justice system.[4]

A case can enter SINAN in two ways. The victim may seek care after an episode of violence, or a health professional may suspect violence during a visit that began for another reason. Suspicion is enough to trigger a notification; confirmation is not required. The notifying network is broad. It includes hospitals and emergency units, but also primary care units, the Family Health Strategy teams that visit households in their area, and psychosocial care centers (CAPS). This network reaches low-income households in particular, since most of them rely exclusively on the public health system (Instituto Brasileiro de Geografia e Estatística, 2020).

SINAN records two dates for each case: the date of the event and the date of notification. We use the date of the event. Our outcomes are daily counts, and we are interested in the day on which the conflict occurs. Notification can take place several days after the event, so using the notification date would introduce measurement error in the timing of cases. This error is not random: notification dates shift weekend cases toward the next working day.[5]

The date of the event has its own measurement problem. When a case is notified long after it occurred, the exact date is often unknown, and the first day of the month is recorded instead. This heaping on day one increases with the delay between the event and its notification.[6] Table A.1 in the Appendix reports the distribution of cases by the delay between

[4]Since March 2020, health services must also report suspected or confirmed cases of violence against women to the police within 24 hours (Law 13,931/2019).

[5]With the notification date, Monday concentrates 19.1 percent of cases and Sunday 10.6 percent. With the date of the event, Sunday is the most frequent day (16.7 percent).

[6]Among cases notified on the same day they occurred, about half of all notifications, the first day of the

occurrence and notification, together with the excess of cases on the first day of the month. We adopt a conservative approach and count only cases notified within seven days of the event. We also drop January from the sample, since heaping in that month remains high even after this restriction.[7]

We aggregate cases into daily counts at the municipality level. Our main outcome is the number of cases of interpersonal violence, which we split by the sex and age of the victim: women and men aged 15 or older, and girls and boys under 15. For women aged 15 or older, we also count cases by type of violence (physical, psychological, financial, and sexual), cases of repeated violence, cases of intimate partner violence, and cases in which the perpetrator was under the influence of alcohol. We compute the same outcomes restricted to cases with a male perpetrator. For children under 15, we count cases by type of violence, including neglect, and cases in which the perpetrator was suspected of alcohol use.

Table 1 describes the cases of interpersonal violence in our sample. Between February 2022 and June 2024, the sample municipalities recorded 629,487 cases, or 153 per 100,000 inhabitants per year. Seven in ten victims are women, and the incidence among women is more than twice that among men (208 and 94 cases per 100,000). Most cases involve physical violence (65 percent) and take place in the victim's home (61 percent). The profile of cases differs by the sex of the victim. Among women, 39 percent of cases are perpetrated by a current or former partner, and 41 percent involve repeated violence. Among men, 39 percent of victims are under 15, and the most frequent perpetrators are family members (39 percent); only 10 percent of cases involve a partner. On an average day, a municipality in our sample records 0.26 cases of interpersonal violence, 0.18 of them against women.

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month shows no excess of cases.

[7]The excess of cases in January likely combines two sources. First, violence may genuinely increase around New Year's celebrations, when alcohol consumption and risky behavior rise. Second, when the date of the event is missing or unknown, it may be recorded as the first day of the year. Because we cannot separate the two, we exclude the month.

Table 1: Interpersonal Violence Cases: Descriptive Statistics

	All victims		Female victims		Male victims	
	N	%	N	%	N	%
A. Cases and incidence						
Interpersonal violence cases	629,487		442,328		186,931	
Cases per 100,000 inhabitants per year	152.6		207.7		93.8	
B. Victim age						
Under 15	176,083	28.0	102,893	23.3	73,119	39.1
15 and older	452,591	71.9	338,811	76.6	113,623	60.8
C. Type of violence (not mutually exclusive)						
Physical	407,150	64.7	290,375	65.6	116,639	62.4
Psychological	161,973	25.7	136,708	30.9	25,208	13.5
Sexual	91,749	14.6	80,778	18.3	10,939	5.9
Financial	14,052	2.2	12,157	2.7	1,893	1.0
Neglect	112,339	17.8	56,282	12.7	55,994	30.0
D. Perpetrator sex						
Male	388,135	61.7	294,467	66.6	93,550	50.0
Female	123,400	19.6	78,296	17.7	45,047	24.1
Both sexes	62,488	9.9	33,525	7.6	28,946	15.5
Unknown	55,464	8.8	36,040	8.1	19,388	10.4
E. Relationship to perpetrator						
Spouse or partner	107,745	17.1	95,681	21.6	12,032	6.4
Boyfriend or girlfriend	23,596	3.7	21,360	4.8	2,226	1.2
Former partner	60,546	9.6	56,024	12.7	4,507	2.4
Family member	174,757	27.8	101,749	23.0	72,933	39.0
Acquaintance	75,498	12.0	49,166	11.1	26,311	14.1
Stranger	71,482	11.4	40,899	9.2	30,551	16.3
Caregiver, employer, institution, or police	13,627	2.2	7,044	1.6	6,578	3.5
Other	60,546	9.6	42,325	9.6	18,206	9.7
Not identified	41,690	6.6	28,080	6.3	13,587	7.3
F. Place of occurrence						
Residence	381,948	60.7	289,138	65.4	92,706	49.6
Public road	92,712	14.7	52,841	11.9	39,826	21.3
Other	154,827	24.6	100,349	22.7	54,399	29.1
G. Other characteristics						
Repeated violence	225,663	35.8	182,130	41.2	43,473	23.3
Perpetrator under the influence of alcohol	175,422	27.9	131,638	29.8	43,722	23.4

Notes: Cases of interpersonal violence (self-inflicted violence excluded) notified to SINAN in the 3,050 sample municipalities, February 2022 to June 2024, excluding January, and notified within seven days of the event. Percentages are computed over the total number of cases in each column. Types of violence are not mutually exclusive. Relationship categories are mutually exclusive and assigned in the following order of priority: former partner, spouse or partner, boyfriend or girlfriend, family member (parent, step-parent, child, or sibling), acquaintance, stranger, and caregiver, employer, institution, or police; "not identified" means no relationship is recorded. Incidence is the number of cases per 100,000 inhabitants of each sex per year, using IBGE population figures. The 228 victims of unknown sex are included only in the first column, and the 813 victims of unknown age (0.1 percent) are not shown in panel B.
Source: SINAN, Ministry of Health.

3.2. Bolsa Família Payments

Data on Bolsa Família payments come from the Transparency Portal (Portal da Transparência), which publishes monthly records of every payment made by the program, including the beneficiary's NIS, municipality, and amount paid. Because each payment goes to one designated recipient per family, our counts measure families rather than individuals. For each municipality and month between January 2022 and June 2024, we count beneficiaries by the last digit of their NIS and match these counts with the official payment calendar, which sets the day on which each digit is paid. We compiled the calendar from the program's official announcements.

We use these data to build two municipal measures. The first, PayAcum_{it}, is the cumulative share of municipality i's beneficiaries who have been paid by day t of the monthly payment cycle. Because the number of beneficiaries assigned to each digit varies across municipalities, it follows a different path in each municipality, even on the same calendar day. The second, Benef_i, is the number of beneficiary families per capita in municipality i, averaged over the months before the game went viral. Population figures come from IBGE. We describe how both measures enter our empirical strategy in Section 4.

One concern is that beneficiaries may not withdraw their benefit on the day it becomes available, in which case PayAcum_{it} would misstate when cash reaches households. To address this, we use withdrawal records from the Transparency Portal, which report the date on which each payment was withdrawn. These records are complete only for January to March 2021, before our sample period, and we aggregate them at the national level by NIS digit. In those months, withdrawals closely follow the payment calendar: by the last day of the payment window, 94 to 98 percent of benefits had already been withdrawn (Figure A.1).

3.3. Online Gambling: Google Trends

We measure the popularity of the game with Google Trends data on monthly searches for "jogo do tigrinho" in Brazil, which Google reports on a scale from 0 to 100 relative to the

peak over the selected period. Searches rose sharply from July 2023, peaked in December 2023, and remained high through mid-2024 (Figure A.2). In our main specification, we use this pattern to date the viral spread of the game in July 2023. In alternative specifications, we replace this binary indicator with the continuous index, rescaled to range from 0 to 1.

3.4. Sample

We restrict the sample to municipalities where both violence and payment timing are measured reliably, using three criteria. First, we drop municipalities that report no cases to SINAN for six or more consecutive months, since a zero in these places may reflect missing reports rather than the absence of violence. Second, we keep municipalities with at least 300 beneficiary families on average. In municipalities with fewer beneficiaries, the share paid on each day depends on the NIS digits of a small number of families and is measured with more noise. Third, we keep municipalities with at least 5,000 inhabitants, where underreporting is less likely than in very small towns, in which victims are more likely to know the health staff or police officers in charge of their case. Section 7 shows that our results hold under alternative sample criteria.

Our final sample includes 3,050 municipalities, which account for 91.2 percent of Brazil's population (Table A.2). Most exclusions come from the first criterion, which drops 36.5 percent of municipalities but only 7.5 percent of the population. The panel runs from February 2022 to June 2024, excluding January of each year, for a total of 2,436,950 municipality-day observations.[8]

[8]We also exclude September 1–10 and November 1–10, 2022, when SINAN data show an interruption in data entry.

4. Empirical Strategy

4.1. Exposure to Payday Liquidity

We ask whether the relationship between Bolsa Família payments and domestic violence changed after the game went viral, and whether this change was larger in municipalities where more households receive the program. Our empirical strategy combines three sources of variation: the timing of Bolsa Família payments within each month, differences in program coverage across municipalities, and the viral spread of the game in July 2023. We define the exposure of municipality i on day t as

$$Exposure_{it} = \text{PayAcum}_{it} \times \text{Benef}_i \times \text{Post}_t, \tag{1}$$

where PayAcum_{it} is the cumulative share of the municipality's beneficiaries paid by day t, Benef_i is the number of beneficiary families per capita, and Post_t equals one from July 2023 onward.

PayAcum_{it} is zero before the first payment day of the month, increases on each payment day as another NIS digit is paid, and reaches one on the last payment day, after which it resets to zero until the next cycle (Figure A.3).[9] It varies for two reasons. The payment window shifts from month to month, with the first payment day ranging from the 9th to the 20th and the last from the 22nd to the 31st (Table A.3 and Figure A.4). In addition, the share of beneficiaries assigned to each digit differs across municipalities, so PayAcum_{it} differs across municipalities on the same calendar day.

This second source of variation arises because our data allow us to reconstruct the actual payment cycle in each municipality, rather than a theoretical one based only on the calendar. Consider July 20, 2023, when beneficiaries whose NIS ends in 1, 2, or 3 had already been paid. A theoretical measure that assumes the same number of beneficiaries per digit would

[9] We keep PayAcum_{it} at one for one additional day after the last payment day, to capture responses to the payment cycle that occur the day after payment rather than immediately.

equal 0.30 in every municipality. Our measure instead reflects the actual share of each municipality's beneficiaries with those digits. A value of 0.33, for example, means that 33 percent of the municipality's beneficiaries had been paid by that day, three percentage points more than the theoretical share.

Benef_i captures how large the liquidity shock of each payment cycle is relative to the municipality's population. In other words, it measures how much the local population relies on Bolsa Família transfers. We fix it at its average before the game went viral, so that it does not respond to the game itself. It ranges from 0.04 beneficiary families per capita at the 10th percentile to 0.23 at the 90th. Post_t equals one from July 2023 onward (Section 3.3).

4.2. Main Specification

We estimate the following equation by ordinary least squares:

$$Y_{it} = \beta \, Exposure_{it} + \alpha_i + \delta_{m(t)} + \lambda_{w(t)} + \theta_{d(t)} + \varepsilon_{it}, \tag{2}$$

where Y_{it} is the number of cases in municipality i on day t, α_i are municipality fixed effects, $\delta_{m(t)}$ are year-by-month fixed effects, $\lambda_{w(t)}$ are day-of-week fixed effects, and $\theta_{d(t)}$ are day-of-month fixed effects. The equation also includes all lower-order interactions of the terms in $Exposure_{it}$; the main effects of Benef_i and Post_t are absorbed by the fixed effects. Our coefficient of interest, β, measures how the relationship between payments and domestic violence changed after the game went viral in municipalities with higher program coverage. We cluster standard errors at the municipality level.

The day-of-month fixed effects are central to our design. Because Bolsa Família payments are concentrated in the second half of the month, comparing paydays with other days would confound the payment cycle with any recurring pattern within the month. With these fixed effects, identification comes only from days whose position in the payment cycle changes

across months. Days that always fall outside the payment window, such as the 2nd through the 8th, have $\text{PayAcum}_{it} = 0$ in every month, so the day-of-month fixed effects absorb them completely. The effect instead comes from days that fall inside the window in some months and outside it in others, and from days that always fall inside the window but at different points of the cycle. The 20th, for example, falls inside the window in every month of our sample, but the average share of beneficiaries already paid on that day ranges from 0.1 to 0.9 across months. Figure A.5 shows, for each day of the month, the share of months in which it falls inside the payment window.

4.3. Identification

Our design relies on two assumptions. First, the NIS-digit composition of each municipality's beneficiaries must be unrelated to its underlying level of violence. Since the last digit of the NIS does not depend on household characteristics, we expect this to hold. Consistent with this, the shares of beneficiaries assigned to each digit do not jointly predict the level or the trend of domestic violence across municipalities (Table A.5). Nor do they predict a broad set of municipal characteristics, including demographic composition, human development, and the program's coverage (Table A.6).

The second assumption is that, absent the game, the relationship between payments and violence would have evolved similarly in municipalities with different levels of coverage. We assess this assumption with an event-study version of Equation (2), in which we replace Post_t with quarter indicators:

$$Y_{it} = \sum_{q \neq 2023Q2} \beta_q \, \text{PayAcum}_{it} \times \text{Benef}_i \times \mathbb{1}[t \in q] + \alpha_i + \delta_{m(t)} + \lambda_{w(t)} + \theta_{d(t)} + \varepsilon_{it}, \qquad (3)$$

which again includes all lower-order interactions. Each β_q measures the relationship between payments, coverage, and violence in quarter q relative to the second quarter of 2023, the last quarter before the game went viral. We test whether the five pre-period coefficients are

jointly zero. Because a joint test can reject when a single quarter is noisy, we also test for a linear trend across the pre-period coefficients.

4.4. Alternative Specifications

We also estimate three alternative specifications that keep Benef_i fixed and change how we measure the payment cycle and the timing of the game. First, we replace Post_t with the continuous Google Trends index, which captures the intensity of interest in the game rather than a single break. Second, we replace PayAcum_{it} with a binary indicator equal to one on days that fall inside the municipality's payment window, and combine it with the Google Trends index. With this indicator, identification comes only from days that fall inside the window in some months and outside it in others, since the day-of-month fixed effects absorb days that always fall inside or always outside the window. Third, we combine the binary payment-window indicator with the binary Post_t, the most conservative of the four combinations.

5. Results

5.1. Main Results

Table 2 reports three coefficients. The first, the triple interaction, is our coefficient of interest. The other two describe the payday effect before the game went viral: PayAcum measures the change in cases over the payment cycle in a municipality with average coverage, and PayAcum $\times$ Benef measures how much larger this change is for each standard deviation of coverage. We present results for all cases of interpersonal violence (columns 1–3) and separately for female (columns 4–6) and male victims (columns 7–9). For each group, we report the total number of cases and the number of cases among minors and adults. We define minors as victims under 15 years of age and adults as victims aged 15 or older.

Focusing first on the coefficient of the triple interaction, in column 1, we find that the

viral spread of the game strengthened the link between Bolsa Família payments and domestic violence in municipalities with higher program coverage. In a municipality with coverage one standard deviation above the mean, the increase in cases over the payment cycle rose by 0.0077 per day after the game went viral, approximately 3 percent of the pre-period mean of 0.238 cases per municipality-day. The corresponding increases are 4.8 percent for minors and 2.6 percent for adults, both statistically significant.

Crucially, when we split the results by the sex of the victim, the increase holds only for female victims. After the game went viral, the effect of Bolsa Família payments on violence against women in municipalities with higher coverage rose by 0.0059 cases per day (column 4), a 3.5 percent increase relative to the mean. The increase is larger for girls, at 6.0 percent, than for adult women, at 2.7 percent (columns 5 and 6). For male victims, the estimates are smaller and none of them is statistically significant at conventional levels (columns 7–9).

Table 2: Bolsa Família Payments and Domestic Violence

	All victims			Female victims			Male victims		
	All	Minors	Adults	All	Minors	Adults	All	Minors	Adults
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
PayAcum × Benef × Post	0.0077**	0.0031***	0.0045**	0.0059***	0.0023***	0.0035**	0.0018	0.0008	0.0009
	(0.0030)	(0.0012)	(0.0023)	(0.0021)	(0.0009)	(0.0016)	(0.0014)	(0.0007)	(0.0010)
PayAcum × Benef	0.0023	-0.0045***	0.0068***	0.0026**	-0.0027***	0.0053***	-0.0003	-0.0018***	0.0015***
	(0.0015)	(0.0010)	(0.0013)	(0.0012)	(0.0007)	(0.0011)	(0.0006)	(0.0005)	(0.0005)
PayAcum	0.0115***	0.0086***	0.0028*	0.0077***	0.0053***	0.0024	0.0038***	0.0033***	0.0005
	(0.0024)	(0.0016)	(0.0016)	(0.0017)	(0.0010)	(0.0015)	(0.0013)	(0.0008)	(0.0008)
Pre-period mean	0.238	0.065	0.173	0.168	0.039	0.129	0.070	0.027	0.043
Effect size (% of mean)	3.2	4.8	2.6	3.5	6.0	2.7	2.6	3.1	2.2
R^2	0.850	0.615	0.835	0.811	0.509	0.790	0.736	0.484	0.714
Observations					2,436,950				
Municipalities					3,050				

Notes: Each column reports the estimates of equation (2) for the daily number of cases in the municipality. PayAcum × Benef × Post is the coefficient β on $Exposure_{it}$. The other two coefficients refer to the period before July 2023. Benef_i is standardized to mean zero and standard deviation one across the sample municipalities, so PayAcum × Benef and β are expressed per standard deviation of Benef_i, and PayAcum refers to a municipality with average coverage. All regressions include municipality, year-by-month, day-of-week, and day-of-month fixed effects. The pre-period mean is the average of the dependent variable between February 2022 and June 2023, excluding January, and the effect size is β as a percentage of this mean. R^2 includes the fixed effects. The sample covers the 3,050 sample municipalities, February 2022 to June 2024, excluding January. Standard errors clustered at the municipality level are in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Minors are victims under 15 years of age; adults are victims aged 15 or older. Victims of unknown sex are included only in columns (1) to (3).
Source: SINAN, Ministry of Health; Portal da Transparência.

Before the game went viral, the payment cycle was already associated with more violence, particularly against children. In a municipality with average coverage, cases against minors rose by 0.0086 per day during the payment cycle, or 13 percent of the mean, with similar increases for girls and boys (columns 2, 5, and 8). For adults, the corresponding change was small, at 1.6 percent (column 3). Notably, this relationship has been the focus of a large literature on the timing of benefit payments. Our estimates are in line with studies that document more domestic violence shortly after benefits are received (Hsu, 2017) and with evidence that payment schedules move domestic abuse and child maltreatment in the same direction (Carr and Packham, 2021), consistent with the frequent co-occurrence of partner violence and child abuse within the same household (Erel and Burman, 1995; Appel and Holden, 1998). The evidence for children is more mixed, however, as tax refunds are followed by fewer reports of child maltreatment (Kovski et al., 2022).

Program coverage, however, shifts this relationship in opposite directions by age. Each additional standard deviation of coverage reduces the payday increase in violence against minors by 0.0045 cases per day, roughly half of the effect at average coverage, and raises the increase among adults by 0.0068 cases per day, or 4.0 percent of the mean. The same pattern holds for female and male victims. To our knowledge, this divergence has not been documented in the cash transfer literature.

These opposite gradients may be consistent with two channels that operate through the same resources. On the one hand, 84 percent of Bolsa Família recipients are women (Table A.4), and transfers controlled by women tend to be spent on children (Rubalcava et al., 2009). By easing financial pressure on parents, each payment should reduce harsh parenting and neglect, as predicted by the family stress model (Conger et al., 1992). Where more households depend on the program, this relief reaches a larger share of families. On the other hand, the same cash in the hands of women may become a target. Models of instrumental violence predict that partners use violence to extract resources shortly after they arrive (Bloch and Rao, 2002; Hsu, 2017). We return to this channel when we examine

the types of violence below.

Taken together, these results show that the game changed the relationship between Bolsa Família payments and domestic violence in different ways by age. Among adult women, it amplified a relationship that already grew with program coverage, increasing the incidence of violent events at home. Among children, both girls and boys, it eroded the protective role that higher coverage used to play. In both cases, the pattern is consistent with gambling making the cash transfer payment a greater source of conflict within the household.

5.2. Types of Violence

Table 3 reports estimates by type of violence for adult female victims. When a case is notified, the health professional records the type of violence (physical, psychological, financial, or sexual), whether it had occurred before, whether the perpetrator was suspected of alcohol use, and the relationship between victim and perpetrator, from which we identify violence by a current or former intimate partner. Before the game went viral, the positive gradient in coverage that we documented for adult women is concentrated in physical and psychological violence, intimate partner violence, and cases in which the perpetrator had been drinking. This pattern is consistent with the extraction channel discussed above, in which partners use violence to gain control over the resources that payments bring into the household.

After the game went viral, the increase is concentrated in physical and intimate partner violence, which rose by 3.9 and 3.5 percent relative to the mean, respectively (columns 1 and 7). The first estimate is significant at the 1 percent level and the second at the 5 percent level, while the estimates for the other types are small and not statistically significant. In Panel B, we restrict the sample to cases with a male perpetrator. The results are similar in sign and magnitude, and both estimates remain significant at the 5 percent level.

The concentration of the effect in physical and intimate partner violence is consistent with the channels discussed in the introduction. Gambling losses may act as emotional cues that trigger violence (Card and Dahl, 2011), deepen the financial stress of the household, or

Table 3: Domestic Violence against Adult Women, by Type of Violence

	Physical (1)	Financial (2)	Psychological (3)	Sexual (4)	Repeated (5)	Alcohol (6)	Partner (7)
A. Adult female victims							
PayAcum × Benef × Post	0.0040***	-0.0002	0.0005	0.0004	0.0009	0.0010	0.0023**
	(0.0014)	(0.0002)	(0.0008)	(0.0005)	(0.0010)	(0.0009)	(0.0010)
PayAcum × Benef	0.0059***	0.0001	0.0014***	-0.0001	0.0016***	0.0043***	0.0040***
	(0.0011)	(0.0001)	(0.0005)	(0.0002)	(0.0006)	(0.0008)	(0.0008)
PayAcum	-0.0005	0.0006**	0.0018**	0.0014***	0.0026**	-0.0019*	0.0006
	(0.0014)	(0.0003)	(0.0009)	(0.0004)	(0.0011)	(0.0010)	(0.0010)
Pre-period mean	0.101	0.004	0.047	0.014	0.058	0.046	0.064
Effect size (% of mean)	3.9	-4.7	1.1	2.9	1.5	2.3	3.5
R^2	0.726	0.399	0.586	0.370	0.706	0.566	0.658
B. Adult female victims, male perpetrator							
PayAcum × Benef × Post	0.0028**	-0.0002	0.0004	0.0004	0.0005	0.0008	0.0021**
	(0.0013)	(0.0002)	(0.0007)	(0.0005)	(0.0008)	(0.0008)	(0.0010)
PayAcum × Benef	0.0049***	0.0001	0.0015***	-0.0001	0.0020***	0.0036***	0.0038***
	(0.0008)	(0.0001)	(0.0004)	(0.0002)	(0.0005)	(0.0006)	(0.0007)
PayAcum	-0.0006	0.0005*	0.0013	0.0013***	0.0017**	-0.0015*	0.0003
	(0.0011)	(0.0003)	(0.0008)	(0.0004)	(0.0008)	(0.0008)	(0.0009)
Pre-period mean	0.075	0.004	0.039	0.013	0.049	0.040	0.061
Effect size (% of mean)	3.7	-5.6	1.1	3.1	1.0	2.1	3.5
R^2	0.682	0.407	0.579	0.357	0.680	0.546	0.651
Observations	2,436,950	2,436,950	2,436,950	2,436,950	2,436,950	2,436,950	2,436,950
Municipalities	3,050	3,050	3,050	3,050	3,050	3,050	3,050

Notes: Each column reports the estimates of equation (2) for the daily number of cases in the municipality. PayAcum × Benef × Post is the coefficient β on $Exposure_{it}$. The other two coefficients refer to the period before July 2023. Benef_i is standardized to mean zero and standard deviation one across the sample municipalities, so PayAcum × Benef and β are expressed per standard deviation of Benef_i, and PayAcum refers to a municipality with average coverage. All regressions include municipality, year-by-month, day-of-week, and day-of-month fixed effects. The pre-period mean is the average of the dependent variable between February 2022 and June 2023, excluding January, and the effect size is β as a percentage of this mean. R^2 includes the fixed effects. The sample covers the 3,050 sample municipalities, February 2022 to June 2024, excluding January. Standard errors clustered at the municipality level are in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Panel A includes female victims aged 15 or older; panel B restricts them to cases in which the perpetrator is male. Types of violence are not mutually exclusive. Repeated means the violence had occurred before; alcohol means the perpetrator was suspected of alcohol use; partner means the perpetrator is a current or former spouse, partner, boyfriend, or girlfriend.
Source: SINAN, Ministry of Health; Portal da Transparência.

spark disputes between partners over how the payment is used. By contrast, the game did not amplify violence involving alcohol, even though this type showed the strongest association with payments before, which suggests that its effect does not operate mainly through drinking.

Table 4 turns to the types of violence against minors. Before the game went viral, every

type of violence against minors shows the pattern documented above: the payment cycle is associated with more violence in a municipality with average coverage (11 to 21 percent of the mean), and higher coverage offsets about half of this increase. We interpret these opposite signs as the result of two competing channels: violence against mothers around payday that spills over to their children, and the financial relief that each payment brings, which is larger where more households depend on the program. After the game went viral, physical violence against minors increased by 6.4 percent relative to the mean, an estimate significant at the 5 percent level (column 1). We also find an increase for sexual violence (5.3 percent) and financial violence, both significant at the 10 percent level. Financial violence against minors is rare, so its small coefficient implies a 39 percent increase relative to the mean, which we interpret with caution. Because the point estimates are positive for all types of violence, we read these results as a further sign of the erosion of the protective role of coverage documented above.

5.3. Event Study

Figure 1 reports the event-study estimates for the main outcomes among adult women, and Figure A.6 in the appendix reports the corresponding estimates for minors. In none of these outcomes do the pre-period coefficients point to differential trends that would undermine the parallel trends assumption. The coefficients before July 2023 are small and close to zero, they are jointly insignificant, and we find no evidence of a linear trend across them. For reasons of space, we present only the main outcomes, but the same conclusion holds for most of the remaining outcomes, including those for which we find no effect after the game went viral.

The timing of the effect is consistent with the spread of the game. Among adult women, the estimates remain close to zero in the two quarters after July 2023 and peak in the first quarter of 2024, a few months after internet searches for the game reached their maximum in December 2023 (Figure A.2). This lag is expected. The surge in searches likely reflects

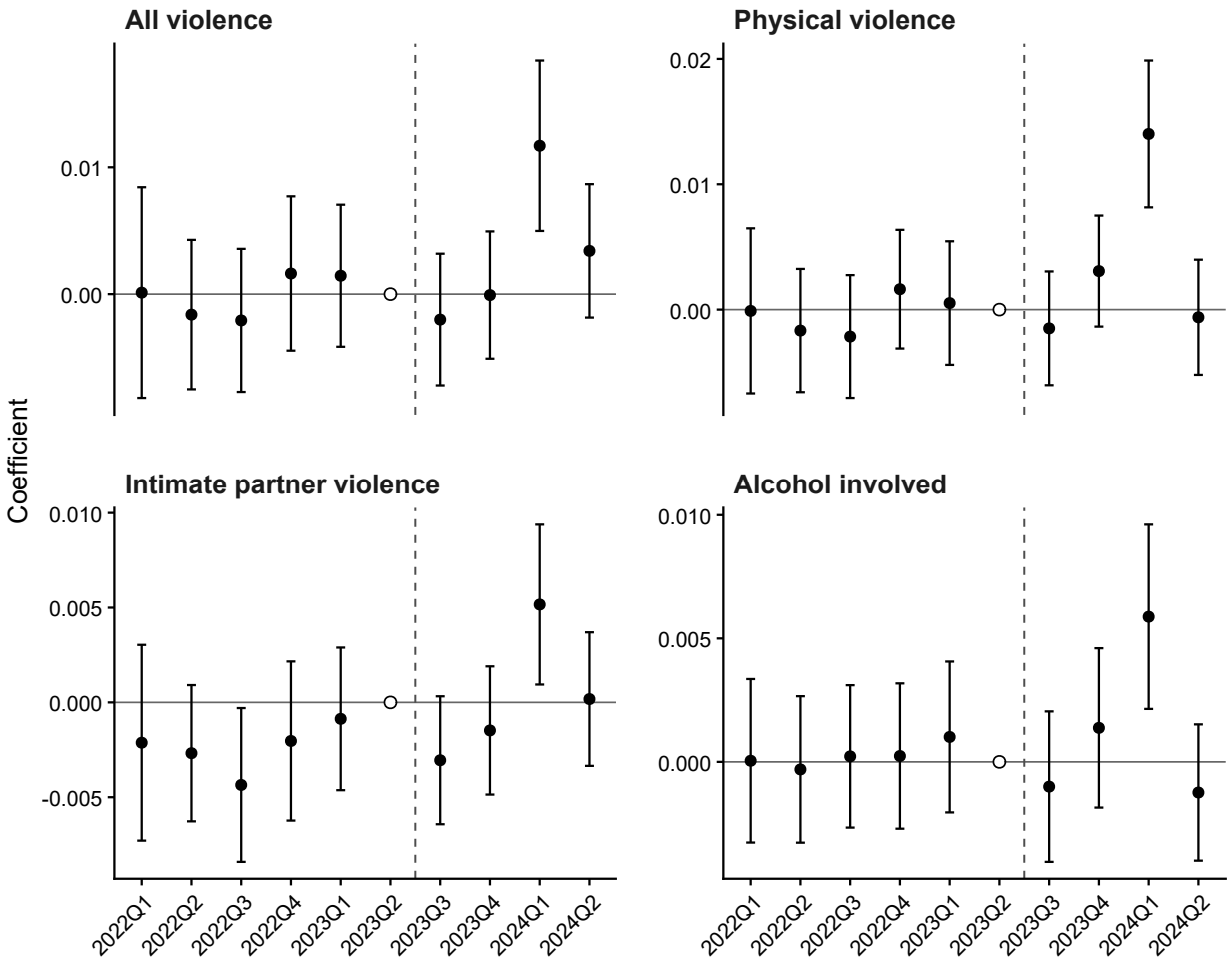
Table 4: Domestic Violence against Minors, by Type of Violence

	Physical (1)	Psychological (2)	Financial (3)	Neglect (4)	Sexual (5)	Alcohol (6)
PayAcum × Benef × Post	0.0011**	0.0007	0.0001*	0.0013	0.0011*	0.0005
	(0.0005)	(0.0005)	(0.0001)	(0.0008)	(0.0006)	(0.0003)
PayAcum × Benef	-0.0009***	-0.0013***	-0.0001**	-0.0017***	-0.0019***	-0.0003
	(0.0003)	(0.0003)	(0.0000)	(0.0006)	(0.0004)	(0.0002)
PayAcum	0.0019***	0.0021***	-0.0000	0.0035***	0.0031***	0.0006
	(0.0005)	(0.0005)	(0.0001)	(0.0009)	(0.0007)	(0.0004)
Pre-period mean	0.017	0.010	0.000	0.030	0.020	0.007
Effect size (% of mean)	6.4	7.4	38.6	4.3	5.3	7.5
R^2	0.297	0.188	0.013	0.549	0.341	0.204
Observations	2,436,950	2,436,950	2,436,950	2,436,950	2,436,950	2,436,950
Municipalities	3,050	3,050	3,050	3,050	3,050	3,050

Notes: Each column reports the estimates of equation (2) for the daily number of cases in the municipality. PayAcum × Benef × Post is the coefficient β on $Exposure_{it}$. The other two coefficients refer to the period before July 2023. Benef_i is standardized to mean zero and standard deviation one across the sample municipalities, so PayAcum × Benef and β are expressed per standard deviation of Benef_i, and PayAcum refers to a municipality with average coverage. All regressions include municipality, year-by-month, day-of-week, and day-of-month fixed effects. The pre-period mean is the average of the dependent variable between February 2022 and June 2023, excluding January, and the effect size is β as a percentage of this mean. R^2 includes the fixed effects. The sample covers the 3,050 sample municipalities, February 2022 to June 2024, excluding January. Standard errors clustered at the municipality level are in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. The sample includes victims of both sexes under 15 years of age. Types of violence are not mutually exclusive. Alcohol means the perpetrator was suspected of alcohol use.
Source: SINAN, Ministry of Health; Portal da Transparência.

people first trying the game, whereas domestic violence should become visible only once gambling becomes a regular activity and its losses accumulate, rather than during casual first contacts. The estimates fall back toward zero in the second quarter of 2024, which indicates that the increase is concentrated in the months following the peak of the game's popularity.

Figure 1: Event Study: Domestic Violence against Adult Women



Notes: Each panel reports the estimates of β_q in equation (3) for the daily number of cases against female victims aged 15 or older in the municipality, for the type of violence indicated in the panel title. The second quarter of 2023 is the reference period (hollow marker), and the dashed line marks the start of the post-period in July 2023. Benef_i is standardized to mean zero and standard deviation one across the sample municipalities, so coefficients are expressed per standard deviation of coverage. Bars show 95 percent confidence intervals based on standard errors clustered at the municipality level. January is excluded from every year, so the first quarter of 2022 covers February and March. The sample covers the 3,050 sample municipalities, February 2022 to June 2024. Alcohol involved means that the perpetrator was suspected of alcohol use. *Source:* SINAN, Ministry of Health; Portal da Transparência.

6. Heterogeneity

Table 5 examines who perpetrates the additional violence. Panel A splits all cases against adult women by the victim's relationship to the perpetrator. The increase after the game went viral is concentrated in violence by a spouse or partner, the only group with a statisti-

cally significant estimate. This group accounts for about half of the total effect but only 28 percent of cases. In contrast, the estimates for boyfriends, former partners, family members, and other perpetrators are small and not statistically significant.

Table 5: Domestic Violence against Adult Women, by Perpetrator

	Perpetrator				
	Spouse or partner	Boyfriend	Former partner	Family member	Other
	(1)	(2)	(3)	(4)	(5)
A. All violence against adult women					
PayAcum × Benef × Post	0.0018**	-0.0002	0.0007	0.0001	0.0011
	(0.0007)	(0.0003)	(0.0005)	(0.0005)	(0.0008)
Mean of dep. variable	0.039	0.007	0.023	0.018	0.052
Effect size (% of mean)	4.6	-3.1	3.1	0.8	2.1
Share of total effect (%)	51	-7	20	4	31
Share of cases (%)	28	5	16	13	37
R^2	0.536	0.222	0.456	0.459	0.644
B. Intimate partner violence against adult women, male perpetrator					
PayAcum × Benef × Post	0.0014**	-0.0000	0.0007	–	–
	(0.0007)	(0.0003)	(0.0005)		
Mean of dep. variable	0.037	0.007	0.021	–	–
Effect size (% of mean)	3.9	-0.2	3.2	–	–
Share of total effect (%)	68	-1	33	–	–
Share of cases (%)	57	10	33	–	–
R^2	0.527	0.213	0.447	–	–
Observations	2,436,950	2,436,950	2,436,950	2,436,950	2,436,950
Municipalities	3,050	3,050	3,050	3,050	3,050

Notes: Each column reports the estimates of equation (2) for the daily number of cases against female victims aged 15 or older in the municipality, split by the victim's relationship to the perpetrator. Panel A includes all types of violence. Panel B restricts cases to intimate partner violence by a male perpetrator, that is, cases in which the perpetrator is a male current or former spouse, partner, or boyfriend, so the columns for family members and other perpetrators do not apply. The relationship to the perpetrator is assigned in the following order of priority when more than one is recorded: former spouse or boyfriend, spouse or partner, boyfriend, family member (parent, step-parent, child, or sibling), and all other perpetrators, including acquaintances, strangers, caregivers, employers, institutions, and the police, or no relationship recorded. Because the groups are mutually exclusive, the coefficients in each panel add up to those for all cases, reported in Table 2, column 6 (panel A), and Table 3, panel B, column 6 (panel B). The share of total effect is β for each group divided by β for all cases, and the share of cases is the mean of the dependent variable for each group divided by the mean for all cases. The coefficients, fixed effects, sample, and standard errors are defined as in Table 2. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.
Source: SINAN, Ministry of Health; Portal da Transparência.

Panel B restricts the sample to intimate partner violence by a male perpetrator. Within

this group, the increase comes from current spouses or partners, who account for two-thirds of the effect. The estimate for former partners is not statistically significant, and the estimate for boyfriends is close to zero. This result is robust to stricter and broader definitions of a current partner based on the victim's marital status (Table A.8). Since most spouses or partners recorded in SINAN live with the victim, this pattern points to conflict between partners who share a home rather than to violence in romantic relationships in general.

The place of occurrence reinforces this interpretation. Table A.7 splits physical violence against adult women by both the perpetrator and the place where the event occurred. Violence by a spouse or partner increases at home, accounting for about a third of the total effect on physical violence, while the estimates for the same perpetrators in public roads or other places are small. Violence by family members does not increase in any location. The only other significant increase, which is small, corresponds to boyfriends and former partners on public roads. Taken together, these results indicate that the effect of the game operates mainly through conflict between cohabiting partners at home, a condition shared by all the mechanisms discussed in the introduction.

Table 6 turns to violence against minors. As for adult women, the increase after the game went viral takes place at home. Violence by family members at home, whether parents, step-parents, or siblings, accounts for about 70 percent of the effect on all violence against minors and for more than three-quarters of the effect on physical violence, well above its share of cases. Cases at school or in other places do not increase. Sexual violence is the exception: its increase comes mainly from perpetrators outside the immediate family, in cases that also occur at home.

Table A.9 breaks these results down further. Among family members, the increase is not driven by a single relative: it is spread across mothers, fathers, and both parents roughly in proportion to their share of cases, and it falls mainly on male perpetrators. For sexual violence, step-parents and other perpetrators stand out. The results for children thus are aligned with those for adult women. In both cases, the additional violence appears where the

Table 6: Domestic Violence against Minors, by Perpetrator and Place of Occurrence

	At home			
	Family member	Other perpetrator	School	Other places
	(1)	(2)	(3)	(4)
A. All violence against minors				
PayAcum × Benef × Post	0.0022***	0.0007*	0.0001	0.0001
	(0.0008)	(0.0004)	(0.0003)	(0.0005)
Mean of dep. variable	0.031	0.013	0.005	0.023
Effect size (% of mean)	7.1	5.7	2.7	0.2
Share of total effect (%)	71	23	4	2
Share of cases (%)	43	18	7	32
R^2	0.504	0.224	0.162	0.443
B. Physical violence against minors				
PayAcum × Benef × Post	0.0008**	-0.0000	0.0002	0.0002
	(0.0003)	(0.0002)	(0.0002)	(0.0002)
Mean of dep. variable	0.008	0.003	0.002	0.005
Effect size (% of mean)	10.1	-1.4	6.6	3.0
Share of total effect (%)	76	-4	14	14
Share of cases (%)	44	17	12	27
R^2	0.183	0.046	0.098	0.101
C. Sexual violence against minors				
PayAcum × Benef × Post	0.0003	0.0006*	0.0001	0.0002
	(0.0003)	(0.0003)	(0.0002)	(0.0003)
Mean of dep. variable	0.005	0.009	0.001	0.006
Effect size (% of mean)	5.6	6.5	3.6	2.4
Share of total effect (%)	28	53	5	14
Share of cases (%)	25	40	7	29
R^2	0.128	0.173	0.059	0.141
Observations	2,436,950	2,436,950	2,436,950	2,436,950
Municipalities	3,050	3,050	3,050	3,050

Notes: Each column reports the estimates of equation (2) for the daily number of cases against victims of both sexes under 15 years of age in the municipality, split by the victim's relationship to the perpetrator and the place of occurrence. Column 1 includes cases at home in which the perpetrator is a family member (a parent, step-parent, or sibling); column 2, cases at home with any other perpetrator or no relationship recorded; column 3, cases at school; and column 4, cases in any other place or with no place recorded. Panel B restricts cases to physical violence and panel C to sexual violence. Because the groups are mutually exclusive, the coefficients in each panel add up to those for all cases, reported in Table 2, column 2 (panel A), and Table 4, columns 1 and 5 (panels B and C). The share of total effect is β for each group divided by β for all cases, and the share of cases is the mean of the dependent variable for each group divided by the mean for all cases. The coefficients, fixed effects, sample, and standard errors are defined as in Table 2. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.
Source: SINAN, Ministry of Health; Portal da Transparência.

payment arrives, inside the household and among its members, and not in schools or public spaces. For adult women, the violence is perpetrated mainly by a male intimate partner, and for children, by close family members.

7. Robustness

In this section, we examine whether our results could be explained by factors other than the spread of the game. We first present placebo tests on the outcome, the distribution of coverage, and the timing of payments, and then assess the robustness of our results to alternative specifications, samples, and measures.

7.1. Placebo Tests

If our estimates reflected a broader change in health outcomes around payment days in high-coverage municipalities, rather than domestic violence, we would expect to find similar patterns in outcomes unrelated to gambling. Such a change could arise, for instance, if the expansion of Bolsa Família in 2023 altered consumption, mobility, or the use of health services at payment time in poorer municipalities. To test this, we estimate equation (2) on outcomes with no plausible link to gambling or household conflict: daily deaths from traffic accidents, other accidents, cancer, diabetes, and respiratory disease, drawn from the Mortality Information System (SIM), and daily notified cases of tuberculosis, leprosy, and leptospirosis, drawn from SINAN. None of these estimates is statistically significant (Table A.10). The same holds for homicides, suicides, and alcohol-related deaths.

A second concern is that our estimates do not depend on which municipalities receive more transfers, or on when the payments take place. If the payday pattern of violence changed after mid-2023 in all municipalities alike, or if our estimates simply reflected chance correlations in the data, the level of coverage of each municipality would not matter, and any random assignment of coverage would yield estimates similar to ours. Likewise, if violence rose after mid-2023 in high-coverage municipalities at any point of the month for reasons

unrelated to payments, the timing of payments would not matter, and a payment window placed elsewhere in the month would also reproduce our estimates. We test each of these possibilities.

First, we keep the actual payment calendar and randomly reassign Benef_i across municipalities 1,000 times. The resulting estimates are centered at zero, and our estimates lie in the upper tail of this distribution: for most of our main outcomes, the two-sided p-values reject that a random assignment of coverage reproduces our estimates. For simplicity, we present these distributions for a subset of our main outcomes in Appendix Figure A.7. Second, we keep the actual coverage and assign each month a fake payment window that starts on a random day between the 2nd and the 10th, a period that almost never overlaps with the actual payment window. In all 500 draws, the estimate based on the fake calendar falls below our estimate, and it is negative on average, as expected when the fake window covers the days before beneficiaries are paid (Figure A.8).

7.2. Robustness Checks

Our specification relies on day-of-month fixed effects that are common to all municipalities. If the pattern of violence within the month differed between high- and low-coverage municipalities for reasons unrelated to Bolsa Família, for instance because wages or other transfers arrive on different days, the interaction between PayAcum and Benef would partly capture these differences, and any change in them after mid-2023 would load on β. In a robustness test, we allow the day-of-month fixed effects to vary with coverage, either linearly or by terciles of Benef_i, and in the most demanding specification we also allow the year-by-month fixed effects to vary with coverage. Our results are robust to all of these alternative specifications. Our estimates are also unchanged when we let the municipality, day-of-week, and day-of-month fixed effects differ before and after July 2023.

A related concern is that economic conditions deteriorated more in high-coverage municipalities after mid-2023, raising financial stress in the household at the same time as the

game spread. If this were the case, controlling for local economic activity, and allowing its effect to differ by coverage and period, should reduce our estimates. We control for monthly net formal job creation (CAGED) and for the number and value of Pix transfers per capita, entered in levels, interacted with Benef_i, and interacted with $\text{Benef}_i \times \text{Post}_t$. Across these specifications, the estimates remain within 7 percent of the main estimates and keep their statistical significance.

The spread of the game coincided with the expansion of Bolsa Família. In June 2023, one month before our post period begins, the program added a new benefit for children and adolescents aged 7 to 18, pregnant women, and nursing mothers, raising transfers to larger families. If larger payments increased conflict in the household at payment time, and more so in high-coverage municipalities, β would capture the expansion of the program rather than the game. Our estimates are virtually unchanged when we control for the average benefit in the municipality and month, either in levels or interacted with an indicator for the months after June 2023.

We also test whether our results are robust to alternative definitions of the sample: we drop municipalities only if they report no cases to SINAN for nine or more consecutive months, lower the minimum number of beneficiary families from 300 to 50, and lower the minimum population from 5,000 to 2,500.[10] Our results do not change. In addition, event study estimates by two-month periods rule out that the peak in the first quarter of 2024 is explained by Carnival festivities in February: the estimates for February are small and not statistically significant, while the effect is concentrated in March and April.

Our results are also robust to how we measure coverage. Using monthly, time-varying coverage instead of the average before July 2023, or coverage in a single pre-period month, April 2023, yields similar estimates. Finally, we estimate the alternative specifications described in Section 4. When both the payment cycle and the timing of the game are measured

[10] We also remove the population threshold while requiring 200 beneficiary families, and raise the minimum number of beneficiary families to 500.

with binary indicators, the estimates remain positive, though smaller and less precise, since the binary payment-window indicator ignores the share of beneficiaries paid on each day and is identified only from the days at the edges of the payment window. Replacing Post_t with the continuous Google Trends index reinforces our results: the estimates remain positive and significant, and several outcomes that are not significant in our main specification become significant. Combining the Google Trends index with the binary payment-window indicator still yields similar estimates.

8. Conclusion

This paper studies how the viral spread of online casino gambling altered the relationship between cash transfer payments and domestic violence in Brazil. We combine the sudden popularity of the jogo do tigrinho in mid-2023 with the Bolsa Família payment calendar, which assigns payment days according to the last digit of each beneficiary's social identification number, and compare violence over the payment cycle before and after the game went viral across municipalities with different program coverage. We find that the game strengthened the link between payments and violence against women and children. In municipalities with coverage one standard deviation above the mean, the increase in cases over the payment cycle rose by 3.5 percent of the pre-period mean for female victims, with a larger increase for girls, while we find no effect for male victims. These results are robust to placebo tests on unrelated outcomes, randomly reassigned coverage, and fake payment calendars, and to alternative specifications and samples.

The profile of the additional violence is consistent with conflict within the household. Among adult women, the increase is concentrated in physical violence and intimate partner violence, is perpetrated by current partners who live with the victim, and takes place at home. Among children, the game eroded the protective role that higher program coverage used to play around paydays. Our data do not allow us to distinguish between the mechanisms that may link gambling to violence, such as emotional cues from losses, financial stress, or disputes

over the use of the transfer. Nor do we observe gambling directly: until 2025, spending on online casinos was invisible even in transaction data, so we rely on Google search intensity to measure the game's popularity. Linking gambling activity to household outcomes at the individual level remains an important task for future research.

Our findings carry lessons beyond Brazil. Cash transfers reach a large share of households in Latin America, and online gambling is expanding rapidly in countries that have only recently regulated it or are still debating how to do so. Our results show that the arrival of an easily accessible form of gambling can turn the liquidity provided by these programs into a source of conflict within the household, with costs that extend beyond gamblers to their partners and children. While we do not evaluate specific policies, this diagnosis supports considering the timing of transfers in the design of gambling regulation, together with prevention efforts targeted at beneficiary households and greater transparency in gambling markets.

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A. Additional Tables and Figures

Table A.1: Heaping on the First Day of the Month, by Notification Delay

Days between event and notification	Cases	% of cases	Cumulative %	Day 1 / days 2–9
0 (same day)	661,587	52.6	52.6	1.03
1–3	303,898	24.2	76.8	1.01
4–7	86,034	6.8	83.6	1.17
8–14	52,204	4.2	87.8	1.39
15–30	46,516	3.7	91.5	1.92
31–90	56,369	4.5	96.0	2.77
91–180	24,036	1.9	97.9	4.51
181–365	18,518	1.5	99.3	6.39
More than 365	8,237	0.7	100.0	8.74
Total	1,257,399	100.0		

Notes: Cases notified to SINAN in the 3,050 sample municipalities with a date of the event between January 2022 and June 2024, including January and self-inflicted violence, grouped by the number of days between the event and its notification. The last column divides the number of cases recorded on the first day of the month by the average number of cases recorded on days 2 to 9. Our analysis sample keeps cases notified within seven days of the event.
Source: SINAN, Ministry of Health.

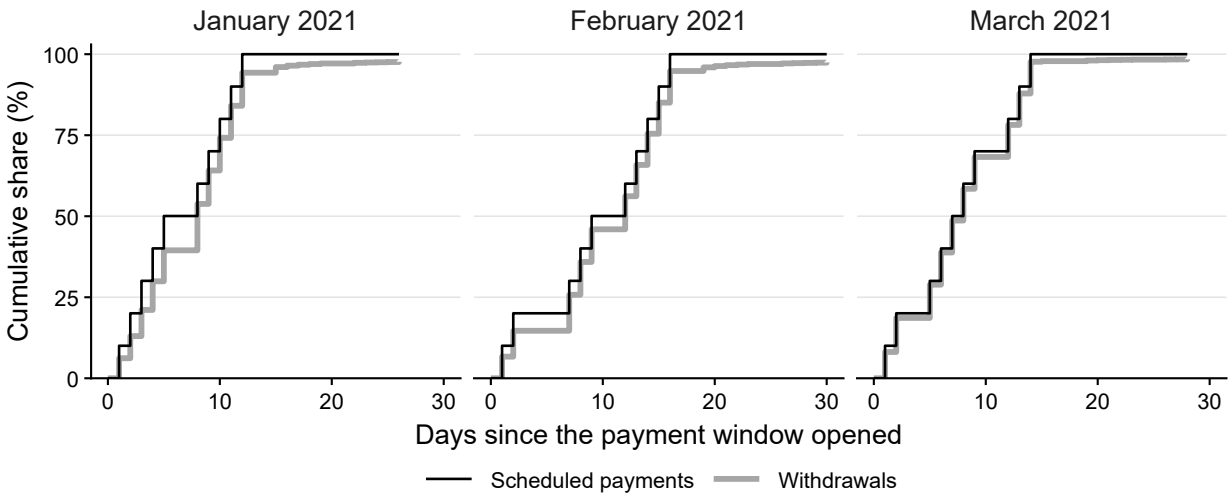


Figure A.1: Scheduled Payments and Withdrawals of Bolsa Família Benefits

Notes: Cumulative share of beneficiaries scheduled to be paid (black) and cumulative share of benefits withdrawn (gray), by day since the payment window opened, January to March 2021. National aggregate by the last digit of the NIS. Withdrawal records are complete only for these months. *Source:* Transparency Portal (Portal da Transparência).

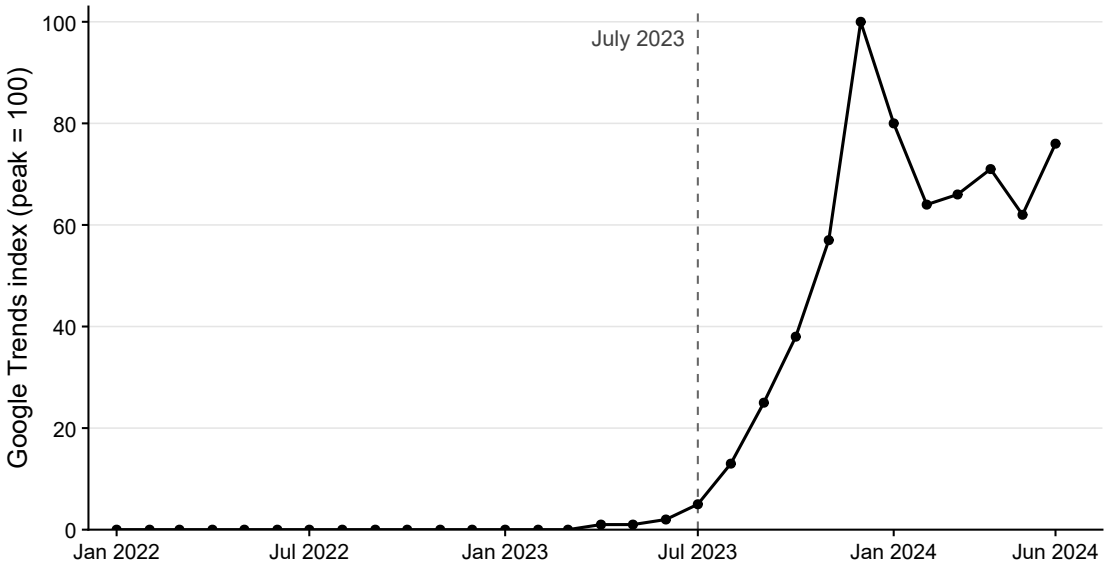


Figure A.2: Google Searches for "Jogo do Tigrinho" in Brazil

Notes: Monthly Google Trends index for searches of "jogo do tigrinho" in Brazil, January 2022 to June 2024, relative to the peak over the period (December 2023 = 100). The dashed line marks July 2023, the month we use to date the viral spread of the game. *Source:* Google Trends.

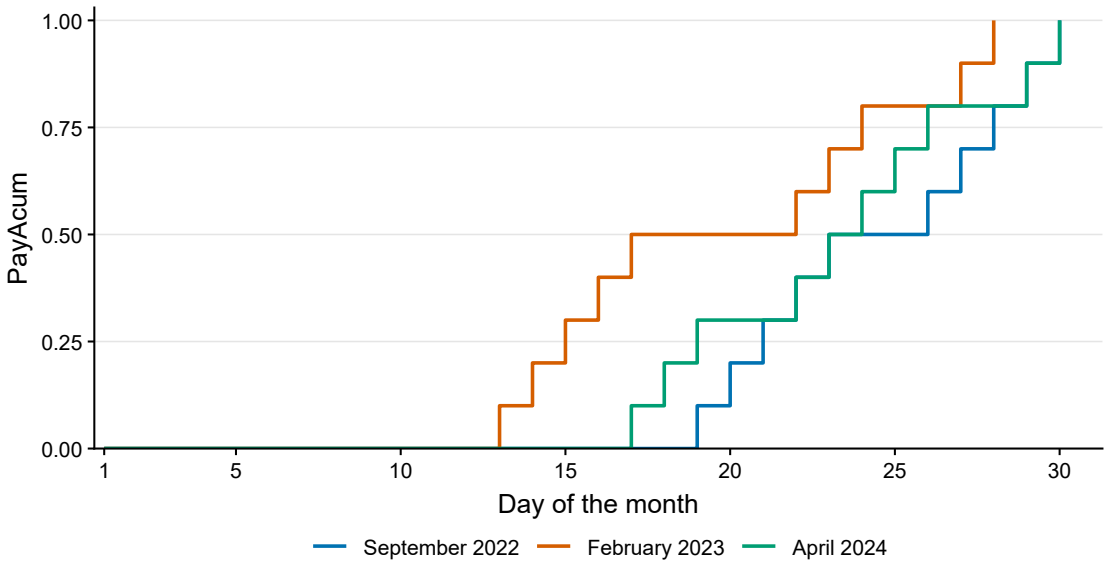


Figure A.3: The Bolsa Família Payment Cycle in Three Sample Months

Notes: Share of the NIS digits already paid by each day of the month in September 2022, February 2023, and April 2024, assuming the same number of beneficiaries per digit. Each step corresponds to the day on which one digit is paid; the share stays constant on weekends and on weekdays with no payment. PayAcum_{it} follows the same schedule but weights each digit by its actual share of the municipality's beneficiaries (Section 4.1). *Source:* Bolsa Família payment calendar.

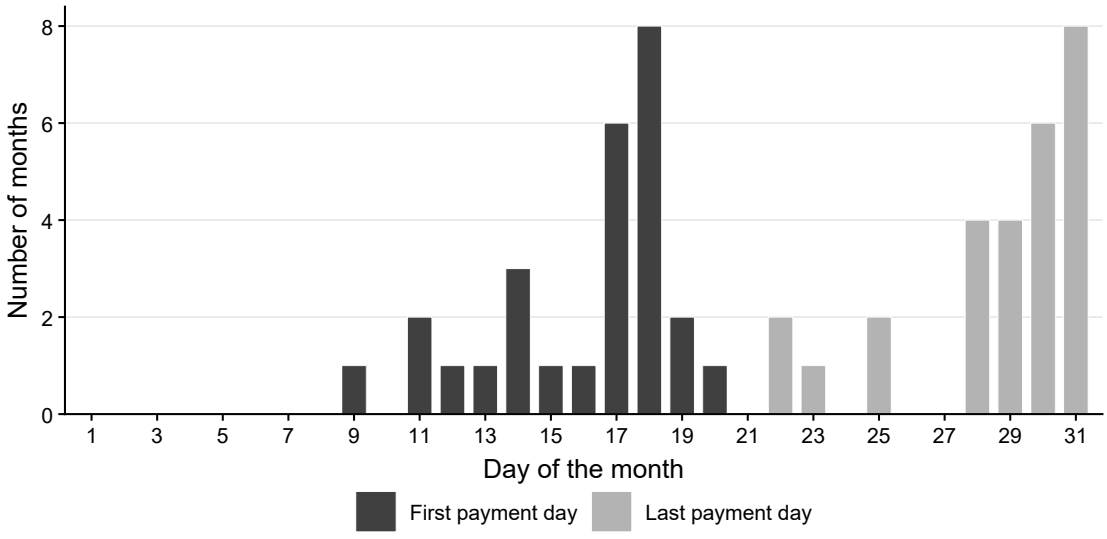


Figure A.4: First and Last Payment Days of the Bolsa Família Cycle

Notes: Number of months in which each day of the month is the first (dark gray) or the last (light gray) payment day of the Bolsa Família cycle, over the 27 months in the sample (February 2022 to June 2024, excluding January). *Source:* Bolsa Família payment calendar.

Table A.2: Sample Construction

	Municipalities		Population	
	N	%	Millions	%
A. Municipalities				
All municipalities in Brazil	5,570	100.0	212.6	100.0
No SINAN reporting gap of six months or more	3,534	63.5	196.6	92.5
At least 300 beneficiary families	3,227	57.9	194.7	91.6
At least 5,000 inhabitants	3,050	54.8	194.0	91.2
B. Days				
February 2022 to June 2024	881			
Excluding January	819			
Excluding SINAN data-entry gaps	799			
Municipality-day observations	2,436,950			

Notes: Panel A applies the sample criteria in sequence. Each row reports the municipalities that remain after the criterion is applied, with their share of all municipalities and of the population of Brazil (IBGE estimates for 2024). The number of beneficiary families is the monthly average between January 2022 and June 2024. Panel B reports the calendar days in the panel; the SINAN data-entry gaps are September 1–10 and November 1–10, 2022.
Source: SINAN, Transparency Portal, and IBGE.

Table A.3: Bolsa Família Payment Calendar

	Mean	Median	SD	Min	Max
First payment day	16.0	17	2.83	9	20
Last payment day	28.6	30	2.80	22	31
Length of payment window (days)	13.6	14	1.28	12	16

Notes: Day of the month on which the Bolsa Família payment window opens (first NIS digit paid) and closes (last NIS digit paid), over the 27 months in the sample (February 2022 to June 2024, excluding January). The length of the window counts calendar days from the first to the last payment day, inclusive.
Source: Bolsa Família payment calendar.

Table A.4: Bolsa Família in the Sample Municipalities

	Mean	SD	P10	Median	P90
A. Coverage					
Beneficiary families per municipality	5,675	20,343	631	2,445	10,580
Benef_i: beneficiary families per capita, pre-period	0.123	0.074	0.037	0.106	0.226
Population in beneficiary families (%)	32.3	19.1	10.2	28.1	58.9
B. Recipients					
Female recipients (%)	84.4	6.0	76.4	85.6	90.9
Recipients whose name is classified by sex (%)	94.5	2.8	91.5	95.1	97.1
C. Average monthly benefit per family (R$)					
Novo Bolsa Família, Mar 2023–Jun 2024	659	28	633	655	686
D. Municipality					
Population (thousands)	61.8	282.2	7.7	21.3	105.5
Municipalities			3,050		

Notes: Distribution across the 3,050 sample municipalities. Variables are municipal averages over the 27 sample months (February 2022 to June 2024, excluding January), except Benef_i, which is averaged over the months before July 2023, as in the regressions. Beneficiary families are counted as payment recipients, one per family. The population in beneficiary families multiplies the number of families by the average family size in the program (2.61 persons); weighted by population, it is 24.0 percent. The sex of recipients is inferred from their first name and is computed over names that can be classified. The average benefit is reported only for Novo Bolsa Família, since the Auxílio Brasil records do not include the complementary benefits paid in 2022.

Source: Transparency Portal (Portal da Transparência) and IBGE.

Table A.5: Balance of the NIS-Digit Composition

	Level		Trend	
	F-statistic	p-value	F-statistic	p-value
Panel A: All Victims				
All cases	0.91	0.514	1.14	0.332
Victims under 15	0.55	0.840	0.76	0.658
Victims 15 and older	1.01	0.427	1.07	0.384
Female victims	0.93	0.494	0.98	0.456
Female victims under 15	0.62	0.780	1.19	0.299
Female victims 15 and older	0.93	0.494	0.96	0.475
Male victims	1.14	0.327	1.22	0.276
Male victims under 15	0.53	0.852	1.10	0.358
Male victims 15 and older	1.31	0.224	1.48	0.150
Panel B: Female Victims				
Physical	0.74	0.674	0.61	0.785
Financial	0.45	0.909	0.79	0.623
Psychological	0.61	0.789	0.82	0.595
Repeated	0.80	0.616	0.85	0.574
Alcohol-related	1.07	0.385	0.63	0.770
Intimate partner	0.87	0.552	0.98	0.453
Panel C: Male Aggressor				
Physical	0.87	0.555	1.05	0.396
Financial	0.65	0.752	0.85	0.566
Psychological	0.52	0.861	0.94	0.486
Repeated	0.72	0.689	1.09	0.368
Alcohol-related	0.85	0.571	1.21	0.282
Intimate partner	0.93	0.495	0.98	0.450
Panel D: Victims Under 15				
Physical	1.25	0.257	1.25	0.258
Psychological	0.44	0.916	1.15	0.326
Financial	1.05	0.397	1.20	0.290
Neglect	0.80	0.617	0.70	0.706
Sexual	0.86	0.558	0.55	0.841
Panel E: Self-Inflicted Violence				
All cases	2.26	0.016	1.26	0.255
Under 15	0.92	0.504	0.84	0.577
15 and older	2.24	0.017	1.23	0.271
Female	2.22	0.018	1.24	0.263
Male	1.80	0.063	0.79	0.630
Municipalities	3,045		3,045	

Notes: Each row reports a joint F-test that the shares of the municipality's Bolsa Família beneficiaries whose NIS ends in each digit (nine shares, with digit 0 omitted) are jointly zero, in a cross-municipality regression of the outcome on the shares, the log of population, the share of the population living in urban areas, and the income component of the 2010 Municipal Human Development Index. Standard errors are robust to heteroskedasticity. The level is the annual number of cases per 100,000 inhabitants between February 2022 and June 2023, excluding January; the trend is the rate between February and June 2023 minus the rate between February and June 2022. Digit shares are averaged over the same pre-period. Five sample municipalities lack data on the controls.
Source: SINAN, Ministry of Health; Transparency Portal (Portal da Transparência); IBGE; Atlas of Human Development in Brazil.

Table A.6: Balance of the NIS-Digit Composition: Municipal Characteristics

	Mean	F-statistic	p-value
Panel A: Demographics			
Population (log)	10.17	0.97	0.464
Population density (log)	3.64	1.17	0.312
Urban population (%)	74.3	0.60	0.797
Women (%)	50.3	0.39	0.940
Population under 15 (%)	20.9	1.21	0.283
Population aged 15–39 (%)	37.3	0.65	0.753
Population aged 60 and older (%)	16.0	0.82	0.601
Black or mixed race (%)	58.2	0.90	0.520
Indigenous (%)	1.3	0.89	0.535
Panel B: Development and Economy			
Human Development Index (HDI)	0.663	1.70	0.084
HDI, income	0.647	1.29	0.237
HDI, education	0.564	1.88	0.050
HDI, longevity	0.803	1.56	0.123
GDP per capita (log)	10.14	1.05	0.399
Agriculture in value added (%)	17.5	1.05	0.395
Household income from work (%)	67.5	1.35	0.207
Panel C: Education and Infrastructure			
Adults with college degree (%)	10.0	1.32	0.222
Households with internet access (%)	80.6	0.65	0.751
Households with piped water (%)	73.9	0.44	0.911
Panel D: Region			
South	16.0	0.79	0.629
Northeast	33.1	1.00	0.441
Panel E: Bolsa Família			
Beneficiary families per capita (Benef_i)	0.123	1.15	0.325
Average benefit (R$, March–June 2023)	660	1.05	0.399
Municipalities		3,044	

Notes: Each row reports a joint F-test that the shares of the municipality's Bolsa Família beneficiaries whose NIS ends in each digit (nine shares, with digit 0 omitted) are jointly zero, in a cross-municipality regression of the characteristic on the shares, without controls. Standard errors are robust to heteroskedasticity. The mean is across the sample municipalities. Digit shares and Benef_i are averaged over February 2022 to June 2023, excluding January. The average benefit covers only March to June 2023, since the Auxílio Brasil records do not include the complementary benefits paid before March 2023. One sample municipality lacks data on piped water.

Source: 2022 Demographic Census and population estimates (IBGE); municipal GDP 2021 (IBGE); Atlas of Human Development in Brazil (2010); Transparency Portal (Portal da Transparência).

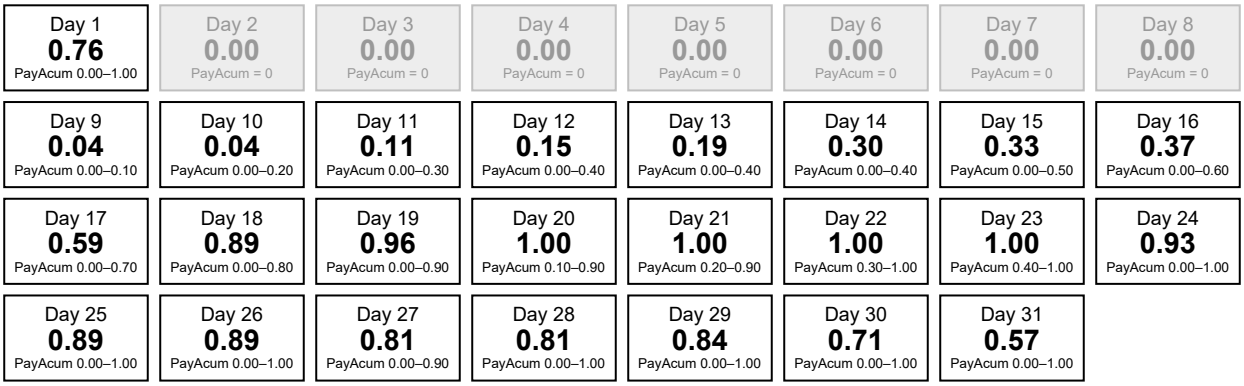


Figure A.5: Which Days Identify the Effect

Notes: For each day of the month, the large number is the share of the 27 sample months in which that day falls inside the Bolsa Família payment window, and the range below it is the minimum and maximum across months of PayAcum_{it}, averaged across the sample municipalities. The window includes the day after the last payment day, so the 1st of the month falls inside it whenever the previous cycle closed on the last day of that month. Days shaded in gray have $\text{PayAcum}_{it} = 0$ in every month and are absorbed by the day-of-month fixed effects. *Source:* Bolsa Família payment calendar and Transparency Portal (Portal da Transparência).

Table A.7: Physical Violence against Adult Women, by Perpetrator and Place of Occurrence

	Spouse or partner			Boyfriend or former partner			Family member		
	Home (1)	Public road (2)	Other (3)	Home (4)	Public road (5)	Other (6)	Home (7)	Public road (8)	Other (9)
PayAcum × Benef × Post	0.0013**	0.0002	0.0002	-0.0004	0.0005**	0.0001	0.0003	0.0001	0.0000
	(0.0006)	(0.0001)	(0.0001)	(0.0004)	(0.0002)	(0.0002)	(0.0003)	(0.0001)	(0.0001)
Mean of dep. variable	0.030	0.002	0.003	0.018	0.004	0.004	0.010	0.0005	0.0010
Effect size (% of mean)	4.2	12.1	8.0	-2.3	14.0	2.0	3.4	17.0	2.0
Share of total effect (%)	32	6	6	-10	13	2	8	2	0
Share of cases (%)	28	2	3	16	3	3	9	0	1
R^2	0.455	0.067	0.095	0.389	0.158	0.118	0.291	0.019	0.035
Observations	2,436,950								
Municipalities	3,050								

Notes: Each column reports the estimates of equation (2) for the daily number of cases of physical violence against female victims aged 15 or older in the municipality, split by the victim's relationship to the perpetrator and by the place of occurrence recorded in SINAN: the residence, a public road, and any other place or no place recorded. Relationship categories are assigned as in Table 5; the second group combines boyfriends and former spouses or boyfriends. Cases with other perpetrators or no relationship recorded are not shown, so the columns do not add up to the total. The share of total effect is β for each group divided by β for all physical violence against adult women (Table 3, column 1), and the share of cases is the mean of the dependent variable for each group divided by the corresponding mean. The coefficients, fixed effects, sample, and standard errors are defined as in Table 2. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Source: SINAN, Ministry of Health; Portal da Transparência.

Table A.8: Intimate Partner Violence by a Current Partner: Alternative Definitions

		Current spouse or partner		
	All	Baseline	Stricter	Broader
	(1)	(2)	(3)	(4)
A. Intimate partner violence against adult women				
PayAcum × Benef × Post	0.0023**	0.0018**	0.0015**	0.0018**
	(0.0010)	(0.0007)	(0.0006)	(0.0007)
Mean of dep. variable	0.069	0.039	0.028	0.040
Effect size (% of mean)	3.3	4.6	5.4	4.4
Share of total effect (%)	100	79	67	78
Share of cases (%)	100	56	41	58
R^2	0.658	0.536	0.474	0.540
B. Intimate partner violence against adult women, male perpetrator				
PayAcum × Benef × Post	0.0021**	0.0014**	0.0014**	0.0014*
	(0.0010)	(0.0007)	(0.0006)	(0.0007)
Mean of dep. variable	0.065	0.037	0.027	0.038
Effect size (% of mean)	3.3	3.9	5.2	3.7
Share of total effect (%)	100	68	67	67
Share of cases (%)	100	57	42	58
R^2	0.651	0.527	0.466	0.532
Observations	2,436,950	2,436,950	2,436,950	2,436,950
Municipalities	3,050	3,050	3,050	3,050

Notes: Each column reports the estimates of equation (2) for the daily number of cases of intimate partner violence against female victims aged 15 or older in the municipality, that is, cases in which the perpetrator is a current or former spouse, partner, boyfriend, or girlfriend. Column 1 includes all cases. Columns 2 to 4 report the cases attributed to a current spouse or partner under three definitions. The baseline definition uses the relationship recorded in SINAN. The stricter definition also requires the victim's marital status to be married or in a consensual union, whether the perpetrator is recorded as spouse or as boyfriend. The broader definition adds boyfriends when the victim's marital status is married or in a consensual union. Panel B restricts cases to male perpetrators. The share of cases is the mean of the dependent variable in each column divided by the mean in column 1. The coefficients, fixed effects, sample, and standard errors are defined as in Table 2. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Source: SINAN, Ministry of Health; Portal da Transparência.

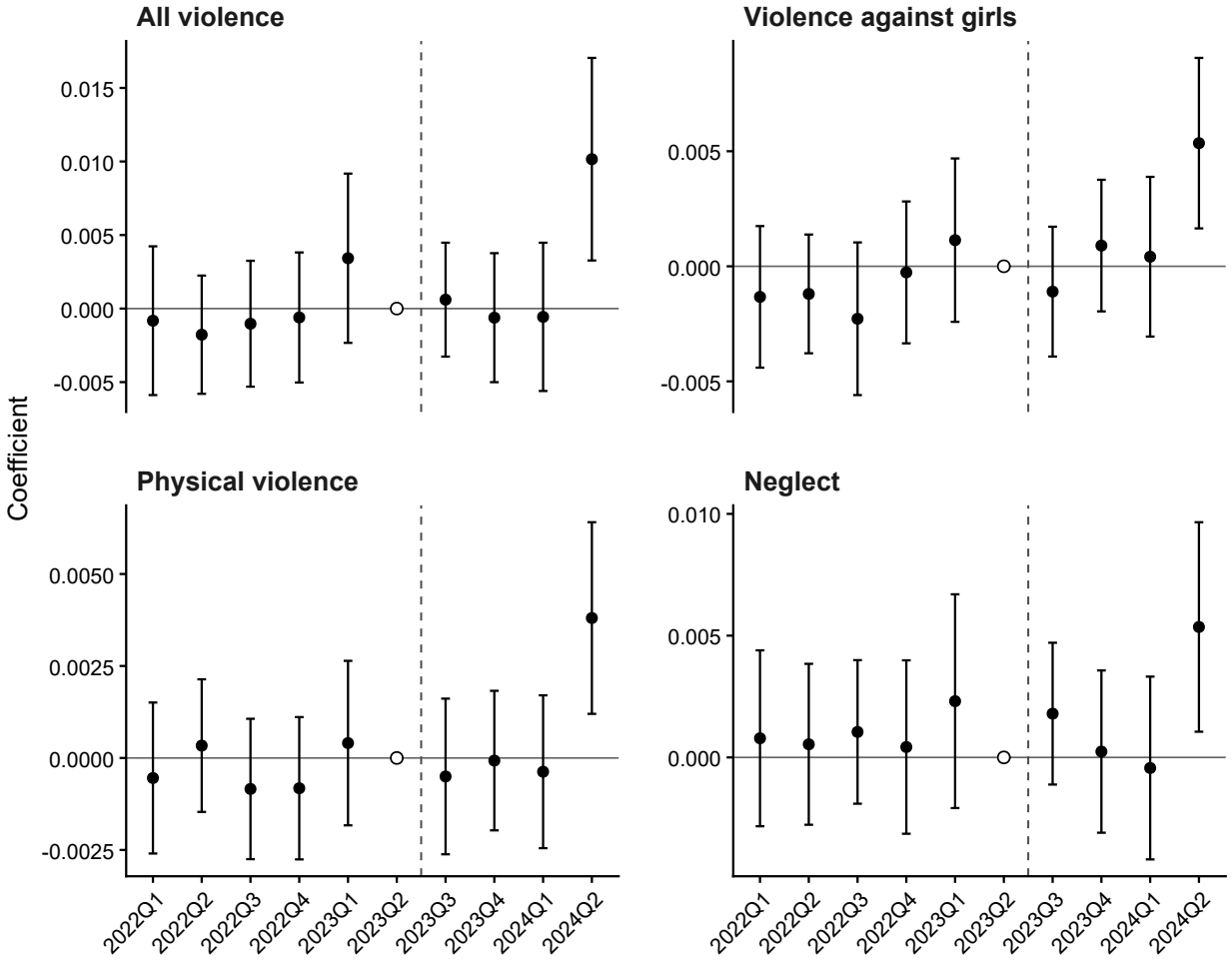
Table A.9: Violence against Minors, by Relationship to and Sex of the Perpetrator

	All violence (1)	Physical (2)	Neglect (3)	Sexual (4)	Share of cases (%) (5)
A. Relationship to the perpetrator					
Both parents	0.0005	-0.0001	0.0005	0.0000	20
	(0.0005)	(0.0001)	(0.0005)	(0.0001)	
Mother	0.0007	0.0003	0.0005	-0.0001	24
	(0.0005)	(0.0003)	(0.0005)	(0.0001)	
Father	0.0005	0.0002	0.0001	0.0001	11
	(0.0003)	(0.0002)	(0.0001)	(0.0002)	
Step-parent	0.0004*	0.0002	-0.0001	0.0003*	5
	(0.0002)	(0.0001)	(0.0001)	(0.0002)	
Sibling	0.0001	0.0001	-0.0000	-0.0000	2
	(0.0001)	(0.0001)	(0.0000)	(0.0001)	
Acquaintance	0.0002	0.0002	0.0000	0.0003	12
	(0.0004)	(0.0002)	(0.0001)	(0.0003)	
Stranger	-0.0000	0.0000	0.0001**	-0.0001	5
	(0.0002)	(0.0001)	(0.0000)	(0.0002)	
Other	0.0007*	-0.0001	0.0002	0.0006**	13
	(0.0004)	(0.0002)	(0.0001)	(0.0003)	
Not recorded	0.0000	0.0001	0.0000	-0.0001	8
	(0.0003)	(0.0001)	(0.0001)	(0.0002)	
B. Sex of the perpetrator					
Male	0.0015**	0.0005	0.0001	0.0010*	40
	(0.0007)	(0.0004)	(0.0002)	(0.0005)	
Female	0.0009	0.0003	0.0006	-0.0000	29
	(0.0006)	(0.0003)	(0.0005)	(0.0001)	
Both	0.0005	0.0001	0.0004	0.0000	22
	(0.0005)	(0.0002)	(0.0004)	(0.0001)	
Not recorded	0.0003	0.0002	0.0002	0.0000	9
	(0.0003)	(0.0001)	(0.0002)	(0.0002)	
Observations			2,436,950		
Municipalities			3,050		

Notes: Each cell reports the estimate of β in equation (2) for the daily number of cases against victims of both sexes under 15 years of age in the municipality, for the group in the row and the type of violence in the column. Panel A groups cases by the victim's relationship to the perpetrator, assigned in the following order of priority when more than one is recorded: both parents, mother, father, step-parent, sibling, acquaintance, stranger, and other perpetrators, including partners, caregivers, employers, institutions, and the police; "not recorded" means that no relationship is recorded. Panel B groups cases by the sex of the perpetrator recorded in SINAN. Within each panel and column, the groups are mutually exclusive, so their coefficients add up to β for all cases (Tables 2 and 4). Column 5 reports the share of all cases against minors in each group. The coefficients, fixed effects, sample, and standard errors are defined as in Table 2. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Source: SINAN, Ministry of Health; Portal da Transparência.

Figure A.6: Event Study: Domestic Violence against Minors



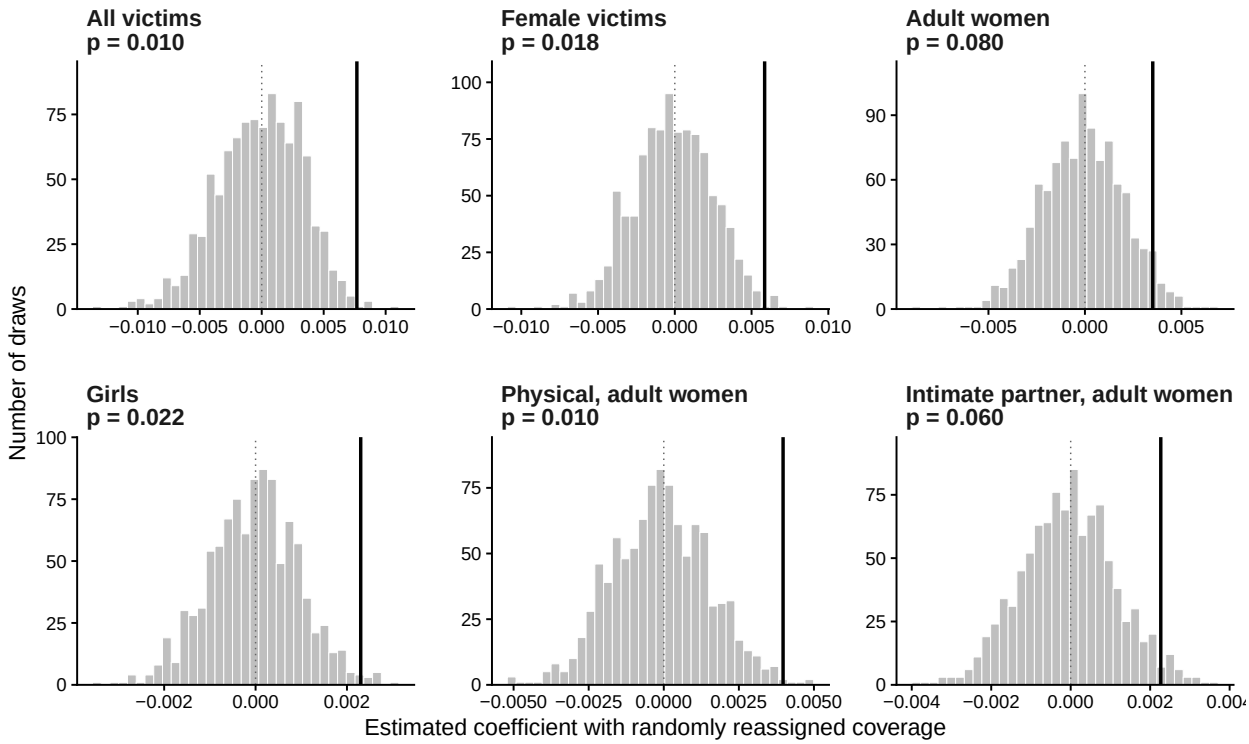
Notes: Each panel reports the estimates of β_q in equation (3) for the daily number of cases against victims under 15 years of age in the municipality, for the type of violence indicated in the panel title. The second quarter of 2023 is the reference period (hollow marker), and the dashed line marks the start of the post-period in July 2023. Benef_i is standardized to mean zero and standard deviation one across the sample municipalities, so coefficients are expressed per standard deviation of coverage. Bars show 95 percent confidence intervals based on standard errors clustered at the municipality level. January is excluded from every year, so the first quarter of 2022 covers February and March. The sample covers the 3,050 sample municipalities, February 2022 to June 2024. Violence against girls restricts cases to female victims; the other panels include victims of both sexes. *Source:* SINAN, Ministry of Health; Portal da Transparência.

Table A.10: Placebo Outcomes

	PayAcum × Benef × Post	SE	Pre-period mean
Panel A. Deaths with no plausible link to gambling (SIM)			
Traffic accident	0.0006	(0.0006)	0.0271
Other accidents	0.0003	(0.0008)	0.0467
Cancer	-0.0005	(0.0016)	0.2032
Diabetes	-0.0005	(0.0008)	0.0585
Respiratory disease	0.0003	(0.0013)	0.1366
Panel B. Notified cases of infectious diseases (SINAN)			
Tuberculosis	-0.0008	(0.0009)	0.0767
Leprosy	0.0002	(0.0005)	0.0183
Leptospirosis	0.0006	(0.0005)	0.0092
Panel C. Deaths that could respond to gambling (SIM)			
Homicide	-0.0004	(0.0006)	0.0363
Suicide	0.0006	(0.0004)	0.0130
Alcohol-related death	-0.0006	(0.0004)	0.0157
Observations	2,436,950		
Municipalities	3,050		

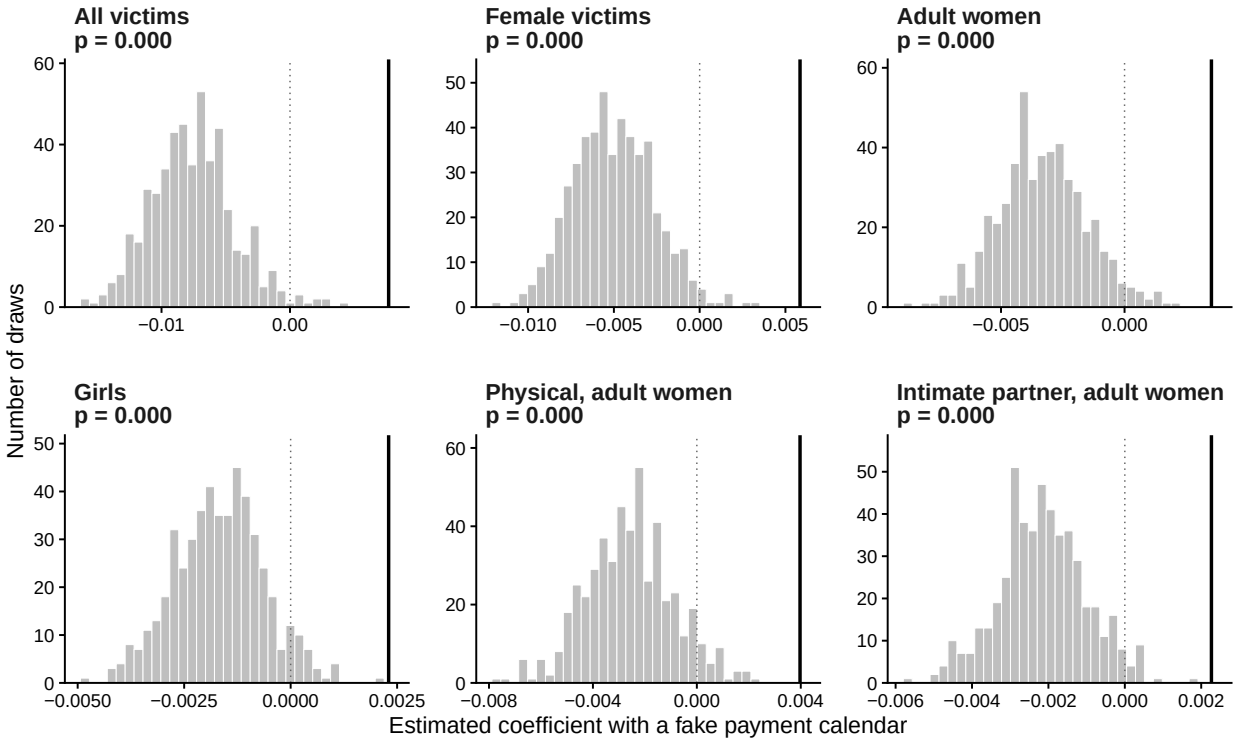
Notes: Each row reports the estimate of β in equation (2) for the daily number of deaths (Panels A and C) or notified cases (Panel B) in the municipality, for the cause in the row. Deaths are dated by the date of death. Benef_i is standardized to mean zero and standard deviation one across the sample municipalities. The fixed effects, sample, and standard errors are defined as in Table 2. The pre-period mean is the average number of deaths or cases per municipality-day between February 2022 and June 2023, excluding January. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.
Source: Mortality Information System (SIM) and SINAN, Ministry of Health; Portal da Transparência.

Figure A.7: Placebo: Randomly Reassigned Coverage



Notes: Each panel shows the distribution of the estimates of β in equation (2) across 1,000 draws in which Benef_i is randomly reassigned across the sample municipalities, keeping the actual payment calendar. The solid line marks the estimate with the actual coverage, and the dotted line marks zero. The p-value is two-sided: twice the smaller of the shares of draws above and below the actual estimate. Coefficients are expressed per standard deviation of coverage. Adult women are women aged 15 or older; girls are female victims under 15. *Source:* SINAN, Ministry of Health; Portal da Transparência.

Figure A.8: Placebo: Fake Payment Calendar



Notes: Each panel shows the distribution of the estimates of β in equation (2) across 500 draws in which PayAcum_{it} is replaced by a fake payment cycle, keeping the actual coverage. In each draw and month, the fake payment window starts on a day drawn at random between the 2nd and the 10th, and beneficiaries are paid in the actual order of NIS digits. The solid line marks the estimate with the actual calendar, and the dotted line marks zero. The p-value is the share of draws with an estimate at least as large as the actual one. Coefficients are expressed per standard deviation of coverage. Adult women are women aged 15 or older; girls are female victims under 15. *Source:* SINAN, Ministry of Health; Portal da Transparência.